
MarketSim: Simulating Stock Markets with Large-Scale Generative Agents

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Abstract

Stock markets are one of the most complex systems in the modern world, where prices emerge from billions of decentralized interactions among heterogeneous participants in an ever-evolving information landscape. While high-fidelity simulation is important for understanding market dynamics, existing approaches face a persistent trade-off between structural and behavioral fidelity. To this end, we propose MarketSim, a large-scale stock market simulation framework with generative agents. MarketSim introduces a hierarchical multi-agent architecture that decouples strategic reasoning from high-frequency execution, enabling LLM agents to operate in a nanosecond-resolution, NASDAQ-like continuous double auction market. Building on this, we simulate over 15,000 heterogeneous market participants whose interactions shape and are shaped by an evolving market environment grounded in more than 12k real-world news articles, policy documents, and earnings reports. To evaluate our proposed MarketSim, we develop a comprehensive benchmark that includes stocks from 8 GICS sectors and 3 representative real-world scenarios, along with 5 stylized facts for market complexity and 5 price-related statistical metrics. Extensive experiments demonstrate that MarketSim not only captures key complexity properties of real-world markets, but also outperforms state-of-the-art baselines in tracking high-frequency price dynamics with an

average MAPE of 3.48%, providing a scalable testbed for market analysis¹.

1. Introduction

Stock markets are the nerve center of the global economy. Despite being powerful engines of economic growth (Levine, 1997), they are often the source of abrupt and widespread systemic collapses (Farmer & Foley, 2009; McKee & Stuckler, 2020). Throughout history, stock markets have experienced countless instabilities such as black swan events, herd behavior, and volatility clustering, all of which reflect the complexity that governs market behavior. In recent years, market-triggered crises have become more frequent, ranging from global disruptions such as the 2008 financial crisis (Farmer & Foley, 2009), to liquidity breakdowns driven by emergencies (McKee & Stuckler, 2020), to flash crashes triggered by policies like the Liberation Day tariff (Wikipedia, 2025). On the other hand, facing such crises, we are at a loss due to the lack of a high-fidelity stock market simulator that can help us understand and anticipate their dynamics.

However, building such a simulator is a non-trivial task. This is primarily because stock markets are complex adaptive systems, where collective outcomes, *i.e.*, prices, emerge from decentralized, nonlinear interactions among large numbers of heterogeneous agents. To decode this complexity, researchers have made efforts to develop agent-based models (ABMs) that aim to replicate stock markets (Arthur et al., 2018; Byrd et al., 2020; Belcak et al.; Axtell & Farmer, 2025). Among them, the ABIDES platform (Byrd et al., 2020) stands out for enabling thousands of simple agents to trade under the continuous double auction (CDA) mechanism, capturing the high-frequency price dynamics in stock markets. While these models largely preserve structures of stock markets, their agents lack *behavioral fidelity*: driven by pre-defined heuristics or rule-based strategies, such agents fail to capture how real-world market participants perceive, interpret, and respond to information. More importantly, this low fidelity in agent behavior further pre-

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¹<https://github.com/tsinghua-fib-lab/MarketSim/>

vents them from grasping one of the stock market’s core mechanisms, *i.e.*, that price changes arise from the collective responses of participants to new information (Fama, 1970; Axtell & Farmer, 2025).

Recent advances in large language models (LLMs) have shown the potential to improve behavioral fidelity (Park et al., 2023; Gao et al., 2024a; Li et al., 2024b). Some researchers have begun to explore replacing traditional agents with LLM-driven ones in stock market simulations (Yang et al., 2025; Gao et al., 2024b; Zhang et al., 2024). However, as a trade-off for current limited agent designs, they typically oversimplify key structures of stock markets. For example, they adopt turn-based trading schemes that fundamentally violate CDA, thereby distorting the essential price discovery process that defines market behavior. Furthermore, they often simulate a small number of agents with oversimplified labels, such as “aggressive” or “conservative”, misrepresenting the scale of real-world participants and their decision-making processes (Brav et al., 2024; Blume et al., 2017). To sum up, their low *structural fidelity* hinders them from reproducing the complex emergent dynamics of real-world markets.

These two lines of research raise a central question: what constitutes a high-fidelity stock market simulator that captures both behavioral and structural fidelity? We propose MarketSim, a high-fidelity LLM-empowered nano-scale stock Market Simulation framework. MarketSim models institutional investors, which dominate real-world trading volume and exhibit highly complex decision-making processes (Brav et al., 2024; Blume et al., 2017). To this end, we introduce a hierarchical, LLM-empowered multi-agent architecture inspired by the organizational logic of real-world institutions. Within each simulated institution, two agent types cooperate: fund manager agents reason over the information landscape to formulate investment strategies and indicative prices, while trader agents execute these instructions through high-frequency trading. This design enables institutional-level intelligence to operate in NASDAQ-like markets at nanosecond resolution. Beyond internal agent dynamics, we situate agents in a rich, evolving environment with two facets. The first is an endogenous market environment collectively shaped by over 15k institutional and background agents (e.g., retail investors and market makers), whose billions of interactions generate prices, returns, and losses that provide feedback for strategic adaptation. The second is an exogenous informational landscape, comprising more than 12k real-world news articles, policy documents, and corporate financial reports.

To evaluate MarketSim, we design a comprehensive benchmark that covers stocks from 8 GICS Level-1 sectors (e.g., Energy, Information Technology), across three representative real-world scenarios: the Liberation Day tariff shock,

DeepSeek’s market debut, and corporate earnings announcements. Moreover, we incorporate 5 stylized facts that qualitatively characterize well-known market complexity, along with 5 price-related statistical metrics that quantitatively measure the alignment between real-world and simulated stocks. Extensive experiments demonstrate that MarketSim faithfully reproduces all five key facts, suggesting that the simulated market exhibits realistic complexity, ranging from black swan events and herding behavior to short-term uncertainties and long-term regularities. Moreover, MarketSim accurately tracks high-frequency price dynamics observed in real-world markets, as validated by five well-established quantitative metrics with an average MAPE of 3.48%. Ablation studies confirm that removing any designs for behavioral or structural fidelity substantially degrades the system. Overall, our work paves the way for a new generation of high-fidelity stock simulators, offering a scalable computational testbed for studying market dynamics, anticipating potential risks, and stress-testing intervention strategies. Our contributions can be summarized into three folds:

- We propose the first high-fidelity stock market simulation framework with generative agents.
- We design a hierarchical multi-agent architecture, which enables agents to participate in NASDAQ-like high-frequency trading.
- We introduce a comprehensive benchmark for stock market simulations, covering stocks from 8 diverse sectors, 3 representative scenarios, 5 stylized facts for market complexity, and 5 price-related statistical metrics.

2. Related Works

We review three lines of related work: (i) stock markets as complex systems, (ii) traditional agent-based modeling (ABM), and (iii) large model-based simulations.

Stock Market and its Complexity. Classical financial theories such as the Efficient Market Hypothesis posit convergence toward equilibrium (Fama, 1970), yet extensive empirical evidence documents persistent stylized facts, including fat-tailed returns, volatility clustering, bubbles, and crashes (Cont, 2001). These phenomena indicate that stock markets are complex adaptive systems in which macro-level patterns emerge from decentralized micro-level interactions (Arthur, 1995). Central to this complexity is the continuous double auction (CDA), where heterogeneous, boundedly rational agents form adaptive expectations in response to evolving endogenous and exogenous information. Faithful market simulation therefore requires modeling both this structural mechanism and agents’ adaptive behavior.

Traditional Agent-Based Modeling. ABM provides a natural bottom-up approach to studying market complexity (Farmer & Foley, 2009). Early models such as the Santa

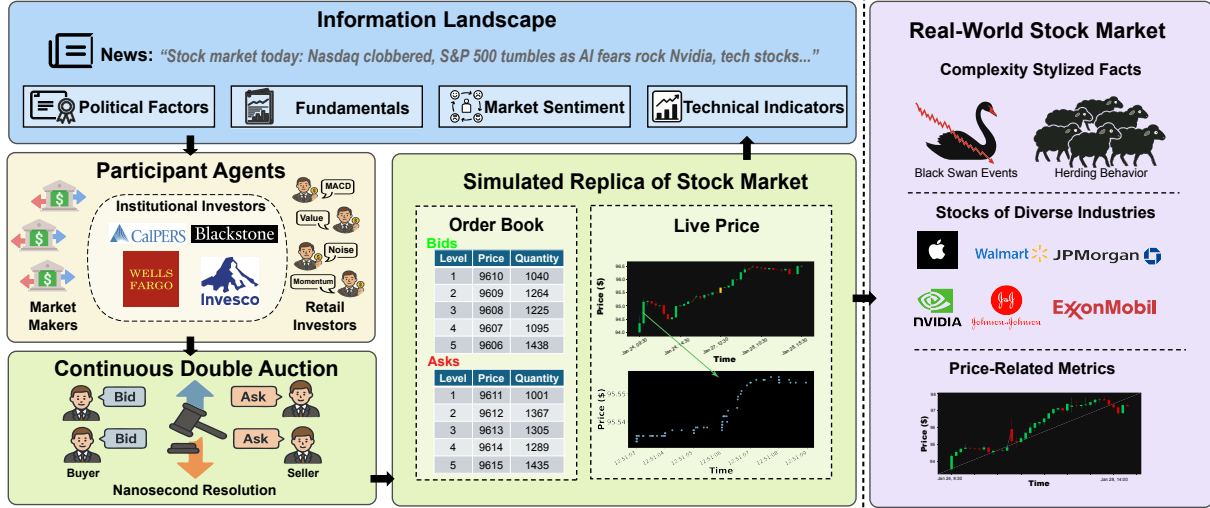


Figure 1. Overview of the proposed framework MarketSim, which captures the complexity of stock markets through three scales.

Fe Artificial Stock Market demonstrated that simple adaptive agents can reproduce key stylized facts (Palmer et al., 1999; Arthur et al., 2018). Subsequent work highlighted the role of agent intelligence in shaping market dynamics (Manahov et al., 2014). More recently, ABIDES significantly advanced structural fidelity by simulating Nasdaq-like markets at scale (Byrd et al., 2020). However, traditional ABMs suffer from limited behavioral fidelity: agents rely primarily on structured order-flow data, ignore rich unstructured information (e.g., news and policy), and employ exogenously specified, simplistic decision rules (Friedman, 2018).

Large Model-Based Simulation. LLMs offer a promising avenue for improving behavioral fidelity by enabling agents to process and reason over complex information (Yu et al., 2024; Xiao et al., 2024). Recent studies integrate LLM-driven agents to generate more human-like market behaviors (Yang et al., 2025; Gao et al., 2024b; Zhang et al., 2024). However, these gains often come at the expense of structural fidelity, as key mechanisms such as the CDA are simplified into turn-based interactions and market scale and heterogeneity are reduced. Data-driven large market models such as MarS simulate order books at scale using a single generative model (Li et al., 2024a), but their black-box nature limits interpretability and the study of emergent phenomena, especially in unprecedented scenarios (Farmer & Foley, 2009).

Summary. Together, these lines of work highlight an unresolved challenge: high-fidelity market simulation requires a unified framework that simultaneously captures **structural and behavioral fidelity**, which is the focus of this work.

3. Framework

To address the challenge of achieving both behavioral and structural fidelity, we introduce MarketSim, a LLM-

empowered stock market simulation framework. As illustrated in Figure 1, the framework is organized into three hierarchical scales: the micro level, which defines the participant agents; the meso level, which delineates the market structures; and the macro level, which constitutes the information landscape. We will elaborate on each of them in the following sections.

3.1. Micro: Participant Agents

The micro level of MarketSim populates the simulation with a heterogeneous agent population with high behavioral fidelity. First, we focus on institutional investors, who not only account for the majority of trading volume but also display complex, information-based decision-making processes in real-world markets (Brav et al., 2024; Blume et al., 2017). To capture their complex reasoning while preserving high-frequency trading capabilities, we design a hierarchical agent architecture empowered by LLMs. Second, we incorporate a rich ecosystem of background agents, such as heuristic-driven retail investors and liquidity-providing market makers.

3.1.1. INSTITUTIONAL AGENTS

Our institutional agent adopts a hierarchical two-level architecture to address a fundamental trade-off. While LLMs offer the deep reasoning capabilities necessary for high behavioral fidelity, they are too slow and computationally intensive for high-frequency market interactions. To resolve this, we draw inspiration from real-world investment institutions, where fund managers make low-frequency, information-rich strategic decisions, and traders focus on executing those strategies at high speed and optimal prices (Golec, 1996; Cohen et al., 2005). This division of labor enables institutions to operate effectively in dynamic markets. Therefore, following this real-world division, we separate the

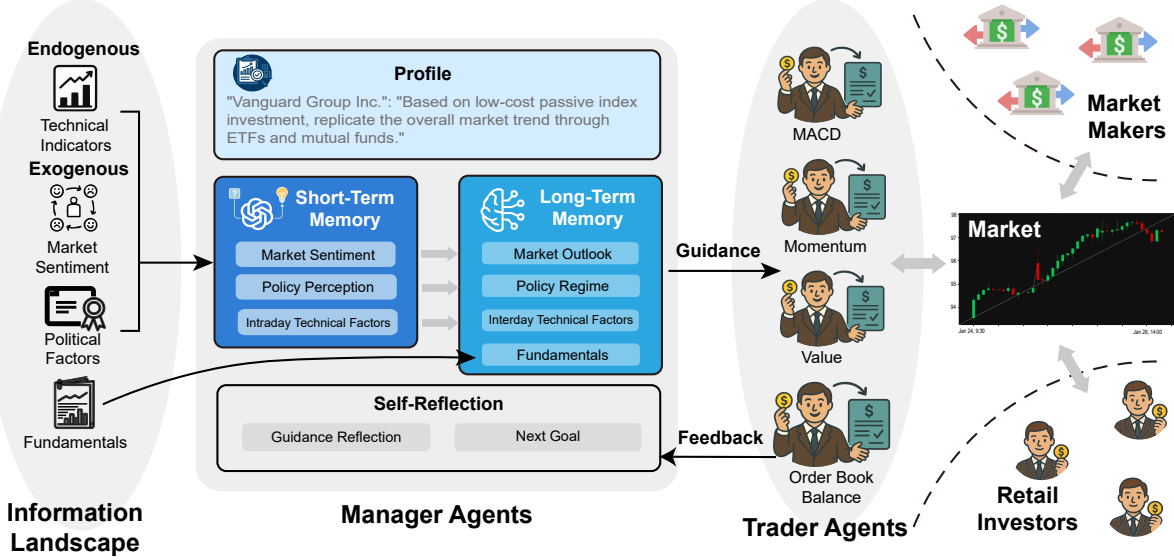


Figure 2. A hierarchical multi-agent architecture for simulating institutional investors.

strategic “brain” from the tactical “hands”: a high-level manager agent, empowered by an LLM, handles complex low-frequency decisions, while a team of low-level trader agents rapidly executes the resulting decisions at nanosecond speed. Formally, the overall of an institution i , denoted as $\pi_{\text{inst}}^{(i)}$, is a composition of its manager’s policy $\pi_{\mathcal{H}}^{(i)}$ and the policies of its K individual traders $\{\pi_{\mathcal{L}}^{(j)}\}_{j=1}^K$.

The Manager Agent $\mathcal{H}$. The manager agent acts as the strategic core of the institution, designed to simulate the cognitive process of a real-world fund manager. As shown in Figure 2, its cognitive architecture comprises four key components: an empirically-grounded profile, dynamic memories encompassing both short- and long-term storage, and accumulated experience.

First, the agent’s cognitive process begins by perceiving the information landscape, drawing on a rich information set I_t from both exogenous sources, such as news and policy signals, and endogenous signals, such as the live order book. This raw data is distilled into short-term memories $\mathcal{M}_{\text{ST},t}$, which capture the agent’s real-time awareness of the present market state. These memories include sentiments derived from external news and policies, as well as technical indicators from market data.

This real-time perception is subsequently consolidated into the agent’s knowledge base, stored as long-term memory $\mathcal{M}_{\text{LT},t}$. This knowledge accumulates through two pathways. Salient short-term memories, e.g., the enduring economic impacts of an initial news shock, are periodically summarized and transferred into long-term memory. In parallel, fundamental information like corporate earnings reports, which indicate a firm’s underlying financial health, is directly encoded into this long-term knowledge base.

Beyond interpreting external data, the agent also learns from its own actions through self-reflection. By evaluating market feedback, e.g., the profitability of past strategies, it updates its experience $\mathcal{E}_t$, enabling continual adaptation of its decision-making policy. Finally, these dynamic cognitive processes are all moderated by the agent’s intrinsic profile. Each manager agent is initialized with a unique, empirically-grounded profile $\mathcal{P}^{(i)}$ that delineates its investment style. This process is formally represented by defining the agent’s comprehensive internal state:

$$A_{H,t}^{(i)} = (\mathcal{P}^{(i)}, \mathcal{M}_{\text{ST},t}^{(i)}, \mathcal{M}_{\text{LT},t}^{(i)}, \mathcal{E}_t^{(i)}). \quad (1)$$

The agent’s policy π_H transforms $A_{H,t}^{(i)}$ into a trading guidance signal G_t :

$$G_t = \pi_H(A_{H,t}^{(i)}) = (\mu_t, \mathbf{w}_t), \quad (2)$$

where μ_t is an indicative price, and $\mathbf{w}_t$ is a vector of weights determining the allocation among different trading strategies for its subordinate trader agents.

The Trader Agents $\mathcal{L}$. The trader agents are the high-frequency execution arm of the institution, representing individual traders who act on the manager’s guidance. They are lightweight, computationally efficient agents that receive the strategic guidance tuple $G_t = (\mu_t, \mathbf{w}_t)$ from their high-level manager agent. Upon receiving this guidance, each trader agent j executes a two-step process: policy selection and parameterized execution.

First, the agent selects its trading policy for the current period. The manager’s weight vector $\mathbf{w}_t$ acts as a probability distribution over a predefined set of available strategies Π_L (e.g., value-based, momentum-based). The trader agent

samples its current strategy, $\pi_{\mathcal{L},t}^{(j)}$, from this categorical distribution:

$$\pi_{\mathcal{L},t}^{(j)} \sim \text{Categorical}(\Pi_L, \mathbf{w}_t). \quad (3)$$

Second, the agent executes its chosen policy in the current period. The indicative price μ_t serves as a key parameter for policies that require a reference price, such as value-based trading. The final action $a_{j,t}$ (e.g., submitting an order) is thus a function of the current market state X_t , conditioned on the parameters derived from the manager’s guidance:

$$a_{j,t} = \pi_{\mathcal{L},t}^{(j)}(X_t; \mu_t). \quad (4)$$

Importantly, the manager’s indicative price is not directly used as the transaction price. It only parameterizes trader-level execution, while actual prices are determined endogenously through order submission, liquidity conditions, and CDA matching. This design mitigates the risk that occasional implausible LLM outputs directly impose price trajectories on the simulated market. Overall, this practice-inspired hierarchical design allows MarketSim to model both the deep, information-driven reasoning of institutions and their rapid, real-time impact on the market.

3.1.2. BACKGROUND AGENTS

To create a realistic market ecosystem for our institutional agents to interact with, we populate the simulation with a diverse population of background agents. While these agents represent a minority of the trading volume, they are crucial for a complete market environment. Here, we focus on two primary categories: retail investors and market makers (Easley & O’Hara, 1995).

Retail Agents. We model retail investors along a spectrum of intelligence. At the simplest level, noise agents emulate the random behavior of uninformed traders; they are activated once per day following a U-quadratic distribution and submit random market orders (Graczyk & Duarte Queiros, 2016). At an intermediate level, momentum agents operate as heuristic-driven trend followers, making decisions based on moving-average indicators derived from high-frequency price data. At the highest level, value agents represent investors conducting pseudo-fundamental analysis, such as inferring value from institutional research reports. Accordingly, we assume they trade based on an estimated fundamental value, computed as the average of indicative prices proposed by all institutional agents, with a variance term added to capture heterogeneity and idiosyncratic noise.

Market Maker Agents. To ensure market liquidity and realistic price dynamics, we include agents that emulate the role of market makers (Easley & O’Hara, 1995). These agents provide liquidity by maintaining both bid and ask orders. They employ an adaptive strategy, dynamically adjusting their quote prices, depths, and spreads in response

to real-time market trading volume and volatility, thereby approximating the behavior of liquidity providers in real financial markets.

3.2. Meso: Market Structures

After defining the micro-level agent population, we specify the meso-level market structure governing their interactions. To ensure high structural fidelity, we model the market as an asynchronous, event-driven system operating under a continuous double auction (CDA), following the design principles of ABIDES and real-world NASDAQ protocols (Byrd et al., 2020). All market interactions are represented as discrete, time-stamped messages and processed in strict chronological order with nanosecond resolution, ensuring causal consistency. Order execution follows a standard lifecycle. An agent submits an order specified by its action, price, quantity, and timestamp. Once activated, the order is processed by a central matching engine that maintains the limit order book (LOB) and matches orders using strict price–time priority. A trade is executed when matching conditions are satisfied, with the execution price determined by the prevailing market price under the CDA mechanism (see Appendix O). Any unfilled quantity is added to the LOB, and the execution outcome is returned to the originating agent.

3.3. Macro: Information Landscape

The macro-level foundation of MarketSim is the information landscape, which underpins all agent decision-making. This landscape comprises two distinct types of information: endogenous information, generated within the simulated market, and exogenous information, sourced from real-world data and events. These two components play complementary roles: endogenous information maintains internal coherence and dynamic feedback within the simulation, while exogenous information anchors agent behavior to external realities, ensuring relevance to actual market narratives and shocks.

Endogenous Information. Endogenous information reflects the real-time internal state of the market, derived primarily from the order book. All agents can query the market structure to access a stream of structured data points, including current bid-ask spreads, market depth, and midpoint prices. This information allows agents, particularly those driven by technical rules, to form perceptions of the market’s immediate liquidity, volatility, and short-term trends.

Exogenous Information. Exogenous information grounds the simulation in real-world scenarios. To this end, we collect and inject a corpus of real-world data aligned with the simulation period, including news articles, major policy announcements, and corporate earnings reports. This rich and often unstructured information is crucial for the LLM-driven manager agents, enabling them to develop nuanced,

human-like perceptions of firms’ fundamental values, relevant political dynamics, and overall market sentiment. As a result, the simulation can respond to the same external events that shape real-world market behavior.

4. Experiments

To evaluate MarketSim, we conduct a series of experiments addressing three research questions (RQs) on realism, accuracy, and generalization, complemented by an ablation study that assesses the contribution of each key component.

- **RQ1. Qualitative Realism:** Can MarketSim reproduce the well-established stylized facts that capture the complexity of real-world stock markets?
- **RQ2. Quantitative Accuracy:** How closely do the price dynamics generated by MarketSim align with real-world data, as measured by a suite of quantitative metrics?
- **RQ3. Generalization:** Can MarketSim generalize across varying market conditions, such as different industrial sectors and diverse types of real-world news events?

Experimental Benchmark

Stocks and Scenarios. We design a comprehensive evaluation benchmark covering a diverse set of stocks and shock scenarios to rigorously test the capabilities of MarketSim. Our stock selection spans eight distinct GICS Level-1 sectors (i.e., Information Technology, Communication Services, Consumer Staples, Healthcare, Financials, Industrials, Energy, and Utilities), ensuring that evaluations are not limited to a single industry. To mitigate the risk of LLM data leakage, we choose three real-world shock events from late 2024 to early 2025: (i) the “Liberation Day” tariff, representing a global policy shock; (ii) DeepSeek’s market debut, reflecting a sentiment-driven shock; and (iii) corporate earnings announcements, capturing fundamental information disclosures. Each scenario is grounded in a rich corpus of real-world data, including over 12k news articles, financial reports, and policy releases, sourced from FintHub, Bloomberg, Newsdata.io, Wind, and FactSet. A key feature of our experimental design is its emphasis on heterogeneity. For each scenario, we deliberately include stocks with varied real-world responses. For example, during the tariff shock, globally exposed firms like Apple are heavily affected, while less globally exposed firms like Johnson & Johnson remain relatively insulated. This setup allows us to assess not only whether MarketSim can reproduce general market trends, but also whether it can capture nuanced, firm-specific dynamics. Our primary large language model is DeepSeek R1, with Qwen3-8b and Llama-3.1-8b used for generalizability experiments. Detailed configurations, including selected stocks and data sources, are provided in Appendix X.

Baselines. To rigorously evaluate the effectiveness of MarketSim, we introduce two classes of baselines: (i) predictive models, including autoregressive methods (Moving Average, ARIMA), traditional machine learning models (Linear Regression, LightGBM), and deep learning models (LSTM, Transformer). These models are trained either on the preceding week of price history (for autoregressive models) or on the preceding week of prices combined with news embeddings (for the higher-capacity models); and (ii) ABM baselines constructed by integrating the same predictive models into the ABIDES framework (Byrd et al., 2020), replacing its exogenous indicative price, which in ABIDES is normally derived from the real-world price series, with model-generated values. This replacement avoids the unfairness of comparing MarketSim’s fully endogenous reasoning with a simulator that relies on real-world price trajectories as external guidance. It is worth noting that this comparison is inherently conservative for MarketSim, as prediction and simulation are fundamentally different tasks. Predictive baselines function as curve-fitting models that treat the market as a black box and optimize numerical forecasts, whereas MarketSim performs mechanism generation by reconstructing the underlying agent interactions that produce realistic price dynamics.

Qualitative Realism via Stylized Facts. To assess the qualitative realism of our simulation (RQ1), we examine whether it reproduces five core stylized facts that characterize the emergent complexity of real financial markets (Cont, 2001). These facts reflect the market’s dual nature of short-term unpredictability and long-term structure: non-stationarity and the absence of linear autocorrelation in returns imply limited predictability from historical prices, while deviations from pure randomness manifest as fat-tailed return distributions and persistent volatility clustering driven by information clustering and collective behaviors such as herding. Over longer horizons, aggregated Gaussianity emerges as return distributions converge toward normality, indicating the growing influence of fundamental forces and arbitrage. We verify these properties using unit root tests, ACF analysis, kurtosis evolution across time scales, GARCH models, and Q-Q plots with kurtosis tests.

Quantitative Accuracy via Statistical Metrics. To quantitatively assess the alignment between the simulated price series (RQ2), we employ a collection of five statistical metrics. We begin by evaluating the direct time-series similarity of prices. We use (i) Root Mean Squared Error (RMSE) and (ii) Mean Absolute Percentage Error (MAPE) to measure point-wise accuracy. To capture morphological similarity, we adopt (iii) Dynamic Time Warping (DTW) Distance, which is robust to temporal shifts and distortions between the two series. Moving beyond the price series itself, we assess the distributional similarity of returns using (iv) Q-Q Correlation, which measures the linear correlation of

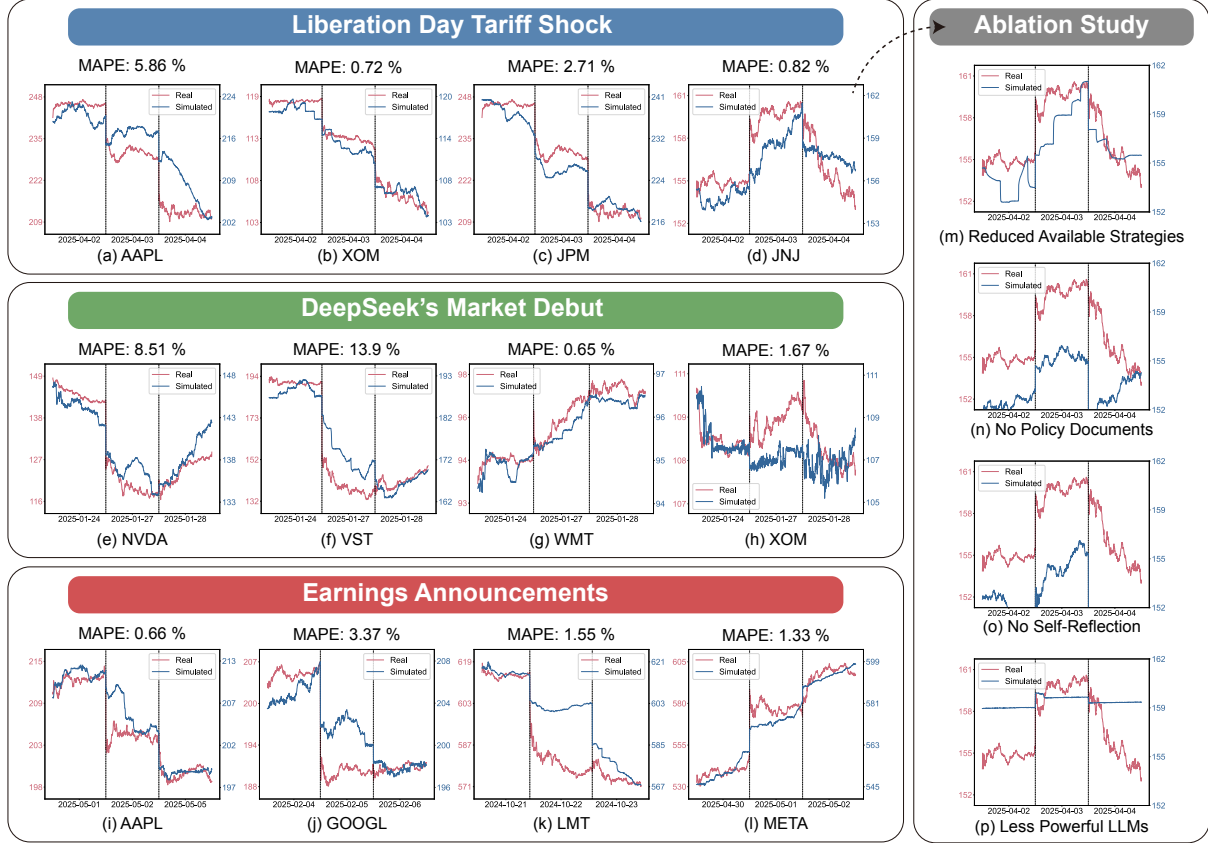


Figure 3. Comparison between real-world and simulated stock data across 12 stocks from 8 GICS sectors and all 3 shock scenarios.

the series’ quantiles. Finally, to evaluate the alignment of volatility characteristics, we use (v) the volatility similarity score, a composite metric that measures similarity across three dimensions: the magnitude of daily price movements, the frequency of significant price changes, and the rate of trend reversals. Details on the evaluation procedures for these stylized facts and quantitative metrics are provided in the Appendix C.

4.1. Experimental Results

RQ1: Qualitative Realism. Our analysis reveals that MarketSim successfully reproduces all five stylized facts across the full range of experiments, covering 12 stocks from 8 GICS sectors and all 3 shock scenarios (summarized in Table S6). This consistent result has two key implications. First, it validates our selected benchmarks, confirming that these stylized facts are indeed universal properties of the empirical data. Second, and more importantly, it demonstrates that by ensuring both structural and behavioral fidelity, MarketSim can capture the emergent complexity of real-world markets—from “black swan” events (Figs. S6-S9, S12-S15, S18-S21b&c) to herding behaviors (Fig. 3), and from short-term uncertainties (Figs. S6-S9, S12-S15, S18-S21a) to

long-term structure. As shown in Tabs. 1 and S10, we find that across all baselines, including predictive models and ABMs, none are able to reproduce the full set of five stylized facts. Each model captures only partial statistical properties and fails to fully capture the complexity in real markets.

RQ2: Quantitative Accuracy. To address RQ2, we assess the alignment between simulated and real-world price series across all scenarios (Tabs. S7, S8, S9). Results show that MarketSim achieves high quantitative accuracy. In terms of time-series similarity, the model attains an average MAPE of 3.48% with consistently low DTW distances, indicating strong morphological alignment. Beyond pointwise accuracy, MarketSim captures deeper statistical properties with high fidelity: Q-Q correlations exceed 0.97 across all twelve experiments, and volatility similarity remains high, confirming accurate reproduction of real-market volatility patterns.

Further analysis shows that accuracy varies with the magnitude of price shocks. Under moderate volatility, such as Johnson & Johnson (JNJ) during the tariff event and Apple (AAPL) during earnings releases, MarketSim performs exceptionally well, with MAPE values as low as 0.82% and 0.66%, respectively. For extreme shocks, such as Vistra

Table 1. Performances of MarketSim and all other baselines on JNJ in Liberation Day tariff shock.

Category		Autoregressive		Traditional ML		Deep Learning		ABM						Ours
Model		MA	ARIMA	Linear	LightGBM	LSTM	Trans.	Linear	MA	ARIMA	LightGBM	LSTM	Trans.	MarketSim
Qual.	Abs. of L.A.	×	×	×	✓	×	×	✓	✓	✓	✓	✓	×	✓
	Fat Tails	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	✓	✓
	Agg. Gauss.	✓	✓	✓	✓	✓	✓	×	✓	×	✓	✓	×	✓
	Volat. Clust.	✓	✓	✓	×	×	✓	×	×	×	✓	×	✓	✓
	Non-Stat.	×	×	✓	×	×	✓	✓	×	×	×	×	✓	✓
Quan.	RMSE	3.833	4.293	1.809	3.808	3.314	6.459	<u>1.816</u>	3.830	4.275	3.792	3.327	6.497	1.614
	MAPE (%)	1.907	2.263	1.067	1.895	1.815	3.246	<u>1.071</u>	1.904	2.250	1.882	1.820	3.264	0.816
	DTW Distance	0.490	0.525	0.291	0.507	0.609	0.153	0.014	0.030	0.030	0.027	0.026	0.005	<u>0.011</u>
	Q-Q Corr.	0.482	0.446	0.892	0.432	0.659	0.817	0.988	0.988	0.989	0.986	0.981	0.995	<u>0.993</u>
	Volatility Sim.	0.012	0.035	0.025	0.247	0.022	0.156	0.433	0.404	0.379	0.464	0.412	<u>0.556</u>	0.796

Corp. (VST) during the DeepSeek debut, which experienced a real-world drop of nearly \$40, the model captures the correct downward trend but underestimates the full magnitude, resulting in higher error (RMSE 23.85, MAPE 13.95%).

Comparisons with all baselines (Tabs. 1, S10) show that MarketSim delivers the most consistent performance across stocks and scenarios. While some baselines perform well on individual metrics, none maintain strong accuracy across the full evaluation suite. MarketSim reduces RMSE by 11%–41% and MAPE by 24%–39% relative to the strongest alternatives, and improves volatility similarity by 15%–43%. Although a transformer-based ABM occasionally attains lower DTW, it performs substantially worse on RMSE, MAPE, and volatility similarity, reflecting shape alignment without realistic reaction timing. By modeling information assimilation, MarketSim generates more temporally coherent price responses.

RQ3: Generalizability. To answer RQ3, we test the framework’s ability to generalize across diverse stocks, sectors, and event types, with results visualized in Fig. 3. The findings confirm that MarketSim successfully captures a wide spectrum of nuanced, firm-specific market reactions. For instance, in response to the single Liberation Day tariff shock, the model captures both the sharp price decline in a trade-exposed firm like Apple (AAPL, Fig. 3a) and the distinct, inverted U-shaped trend of a domestically-focused firm like Johnson & Johnson (JNJ, Fig. 3d). The framework also reproduces other complex, non-linear patterns, such as the U-shaped drop-and-reversal of Nvidia (NVDA, Fig. 3e) during DeepSeek’s market debut. Moreover, it correctly models the behavior of relatively unaffected stocks during the same shock, capturing the steady upward trend of Walmart (WMT, Fig. 3g) and the volatile, sideways consolidation of Exxon-Mobil (XOM, Fig. 3h). By successfully modeling these varied dynamics—from sharp declines to complex reversals and sideways movements—across different industries and under diverse shocks, MarketSim demonstrates robust generalization and the ability to capture the heterogeneous responses that characterize real-world markets.

Ablation Study. To validate our design choices, we conducted a comprehensive suite of 17 ablation experiments. The findings consistently show that reducing the model’s fidelity at either the agent or market level significantly degrades its ability to reproduce realistic market dynamics. First, we confirm the importance of behavioral fidelity. Replacing our empirically-grounded agent profiles with simplistic archetypes like “conservative” or “aggressive” increases MAPE from 0.82% to over 2.8% (Tab. S11). Similarly, degrading the manager’s reasoning ability by using weaker LLMs substantially lowers its performance (Tab. S11), underscoring that sophisticated agent intelligence is crucial. This lack of behavioral fidelity is further highlighted when we remove key cognitive modules; for instance, ablating the self-reflection causes the simulation to fail in reproducing a key stylized fact and increases RMSE by over 10x (Tab. S13). Second, we validate the need for structural fidelity. Restricting the dynamic strategy allocation from the manager (Tab. S12) or adding disruptive market conditions, such as a surge of herd-like individuals or liquidity shocks (Tab. S14), shows that agent behavior is deeply shaped by the surrounding market structure. These results also clarify why MarketSim can reproduce realistic short-term sector responses without per-agent calibration. Realism does not come from matching each simulated agent to an individual real-world counterpart, but from three complementary sources of grounding: empirically initialized institutional profiles at the micro level, CDA-based order-book interactions at the meso level, and a real-world information landscape at the macro level. The ablations show that weakening any of these sources substantially reduces simulation fidelity. Overall, our ablation study confirms a central thesis: the emergent complexity of financial markets, from stylized facts to nuanced price movements, can only be captured when high behavioral and structural fidelity are jointly achieved. In short, fidelity breeds complexity.

Realism of Agent Decisions. To assess the realism of agent decisions, we have conducted a human evaluation study with 18 professional financial practitioners. Each expert evaluates a set of agent-generated decisions along three

dimensions: market consistency, internal coherence, and decision soundness, using a 0–10 rating scale. Results show that the proposed agents receive high average scores across all three dimensions (7.32, 7.44, and 7.15), indicating that practitioners consider the generated decisions realistic and well-reasoned. We further validate this finding by constructing a matched control baseline, where each agent’s decision is paired with a comparable decision from a similar price but unaffected by the shock event. Experts are asked to choose which of the two appears more realistic. Agent decisions are selected significantly more often (a binomial test, $p_0 = 0.745$, $p < 0.001$). Experts then rate both decisions independently along the three dimensions, and agent decisions obtain significantly higher scores in all cases (two-sided Student’s t tests, market consistency: $t = 7.09$, $p < 0.001$; internal coherence: $t = 5.72$, $p < 0.001$; decision soundness: $t = 6.91$, $p < 0.001$). These results show that MarketSim produces agent decisions that are consistently judged by domain specialists as realistic, coherent, and well-grounded.

Applications. We perform two experiments to show the potential of MarketSim as practical testbeds for understanding and anticipating shocks by additionally incorporating (i) 200 momentum-based agents who exhibit trend-chasing behavior, and (ii) agents that submit large-volume orders into the market. We observe that (i) market volatility increases (Fig. S29e), and (ii) liquidity depletion (Fig. S42), both patterns aligning with empirical observations in real-world markets.

5. Conclusion

In this paper, we introduce MarketSim, a simulation framework that resolves the trade-off between behavioral and structural fidelity in stock market modeling. Built on a hierarchical multi-agent architecture, MarketSim reproduces a broad range of complex market dynamics, from emergent stylized facts to firm-specific responses to real-world shocks. Through ablation studies under disruptive conditions, such as liquidity shocks and herd-like trading, we further demonstrate its potential as a scalable testbed for financial risk assessment. Overall, our results provide strong evidence that the emergent properties of stock markets arise from this dual fidelity, underscoring a foundational principle for future research: *fidelity breeds complexity*.

Looking forward, MarketSim opens several directions for future work. First, the current framework focuses on high-fidelity single-stock simulation under Nasdaq-like market microstructure. While cross-asset effects can already enter through shared news, policy signals, and market narratives, future extensions can explicitly model portfolio exposure, correlated behavior, cross-order-book interactions, and systemic contagion across multiple assets. Second, our evaluation focuses on short-horizon market shocks, which are

central to high-frequency crisis dynamics; extending the framework to longer horizons would enable the study of slow institutional adaptation and macro-financial feedback loops. In addition, scaling MarketSim to a broader stock universe mainly requires expanding the supporting data along the same three levels: institutional profiles and trading mandates at the micro level, market-specific trading rules at the meso level, and news, policy signals, and firm fundamentals at the macro level. This modular design allows the core framework to be reused while adapting the data and rules to new stocks and markets. Third, although MarketSim is built around a modular CDA-based design, applying it to other markets, such as A-shares, Hong Kong stocks, or Nikkei, requires adapting market-specific rules, including settlement cycles, tick sizes, trading hours, short-selling constraints, price limits, and volatility-control policies. Finally, improving the numerical robustness, calibration, and auditability of LLM-based market agents remains an important direction.

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Impact Statement

This work investigates LLM-based agents and their use in simulation and decision-making settings. The goal is to improve the methodological understanding of how such agentic systems can be constructed, evaluated, and analyzed in a transparent and controlled manner. The proposed approach may support research on complex social and economic phenomena by enabling safe, low-cost exploration of hypothetical scenarios and interventions. At the same time, we emphasize that the results of simulations should be interpreted with caution and viewed as complementary to, rather than substitutes for, real-world data and human judgment. We also note that high-fidelity market simulators may have dual-use implications, for example by enabling the testing of aggressive or manipulative trading strategies. Therefore, their outputs should be interpreted as research-oriented simulations rather than trading advice, and any real-world use should involve expert oversight and regulatory compliance. This work does not involve real-world deployment or interaction with individuals, and all experiments are conducted in simulated or publicly available environments.

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A. Dataset

Table S2. Data Usage Summary

Usage	Source	Data Type	Data Volume
Initialize Agent Profiles	Finnhub API, Wind, Bloomberg	Stock Price	1.1M data points
		Institutional Ownership	960 holdings
		Financials As Reported	64 reports
		SEC Filings	64 filings
		Social Sentiment	3.2K scores
Provide External Information	Finnhub, Newsdata.io, Dow Jones Factiva	Technical Indicators	39K indicators
		News articles	120K articles
		Policy announcements	

B. Details about Experiments

Table S3. Experimental Scenario Configurations

Scenario	Simulation Period	Selected Stocks
Reciprocal Tariffs	Apr 2-4, 2025	AAPL, JNJ, JPM, XOM
DeepSeek Shock	Jan 24-28, 2025	NVDA, VST, WMT, XOM
Earnings Releases - AAPL 2025Q2	May 1-5, 2025	AAPL
Earnings Releases - META 2025Q1	Apr 30 - May 2, 2025	META
Earnings Releases - GOOGL 2024	Feb 4-6, 2025	GOOGL
Earnings Releases - LMT 2024Q3	Oct 21-23, 2024	LMT

Table S4. Agent Configuration in Simulation Environment

Agent Type	Quantity	Role/Description
Exchange Agent	1	Central market clearing and order matching
Noise Agent	12,000	Random traders simulating market noise
Value Agent	50–100	Fundamental value-based investors
Market Maker Agent	4	Provide liquidity and bid-ask spreads
Momentum Agent	50	Trend-following strategy traders
LLM-driven Manager Agent	10	AI managers guiding trade execution
Trade Agent	2,950	Execution agents under LLM manager guidance

(1) Liberation Day Tariffs (Institutional and Policy Factors): On April 2, 2025, Trump announced his long-promised “reciprocal tariffs” policy, imposing a 10% baseline tax on imports from all countries, with higher rates for nations maintaining trade surpluses with the United States. This policy shock significantly affected global supply chains and multinational corporations. Before the market opened on April 4th, China proposed corresponding countermeasures to U.S. tariffs, further intensifying stock market volatility.

The simulation was conducted from April 2 to April 4. Before the market opened on April 3, we introduced pre-market news concerning the stock together with policy announcements regarding reciprocal tariffs made by former U.S. President Donald Trump. Similarly, before the opening on April 4, we incorporated pre-market news related to the stock along with official announcements from the Chinese government regarding the imposition of retaliatory tariffs. During intraday trading sessions, a selected subset of news items was released in accordance with their actual publication times.

(2) DeepSeek Shock (Market Sentiment and Expectations): In January 2025, Chinese company DeepSeek launched a free AI assistant claiming to use less data at a fraction of incumbent services’ costs. By January 27, the assistant had overtaken ChatGPT in Apple App Store downloads, triggering a massive tech stock sell-off by global investors and causing severe market volatility.

The simulation was conducted from January 24 to January 28. Prior to market opening on January 27, we introduced pre-market news regarding the stock, coinciding with the introduction of DeepSeek into the simulation environment.

(3) Earnings Releases (Fundamental Factors): We select earnings announcement periods for four representative companies (AAPL, META, GOOGL, LMT) to examine market response mechanisms to fundamental information disclosure, including both quarterly and annual reports. For AAPL, META, and LMT, we use quarterly earnings releases as shock events, whereas for GOOGL, we employ its annual report.

Table S5. Scalability guideline for extending MarketSim across stocks and markets.

Level	What is needed	Practical guideline
Micro	Profiles of major trading entities	A basic version can be built from public information on investment style, trading preference, and institutional mandate. A richer version can use professional terminals such as Bloomberg or Wind.
Meso	Market-specific trading rules	For Nasdaq-style simulations, MarketSim can be reused directly. For other markets, one mainly needs to incorporate standardized rules such as tick size, trading hours, short-selling constraints, and price limits.
Macro	News, policy signals, and firm fundamentals	This requires collecting relevant news, policy events, and firm-level fundamentals over the simulated period, from public sources or providers such as Bloomberg, Wind, and Finnhub.

In this simulation scenario, we focus on the release dates of annual and quarterly reports, modeling the trading days both on the announcement date and the adjacent days. Corresponding news items are introduced at the appropriate times to reflect these events.

The proportions of different participant agent types in MarketSim are determined through a hybrid process that integrates empirical evidence, established practices in market simulation, and pilot calibration. Since official data on the exact mix of market participants is not fully disclosed, we approximate the population structure using trading volume and ownership statistics. Reports indicate that retail investors contribute roughly ten percent of trading volume (Adinarayan, 2021), whereas institutional investors hold approximately sixty-eight percent of equity market capitalization (Brav et al., 2024). These observations motivate assigning greater capital and influence to institutional agents within the simulator. We also follow common practice in prior multi-agent simulators such as ABIDES (Byrd et al., 2020), which employ a high proportion of noise agents to provide liquidity.

Building on these empirical and modeling priors, we performed a pilot calibration on a standard trading day prior to the shock. The calibration ensured that the simulated price dynamics exhibit realistic volatility patterns and that the value assessments generated by reasoning-capable agents remain interpretable rather than being dominated by excessive noise trading. The resulting configuration generalizes well across different stocks and shock scenarios without additional tuning, suggesting that it captures a stable approximation of real-world market composition.

Finally, MarketSim operates as an endogenous adaptive system in which the effective activity levels of agents evolve dynamically. Agents with reasoning capabilities may reduce participation under high uncertainty and increase participation when opportunities arise. This adaptive behavior acts as a self-regulating mechanism, making the overall dynamics less sensitive to small variations in the initial population ratios and supporting consistent behavior across a range of market environments.

C. Scalability Guideline

MarketSim is designed as a modular and data-driven simulation framework. Therefore, scaling the framework to a broader stock universe or adapting it to new markets mainly requires expanding the supporting data and market-specific configurations, rather than redesigning the overall system. In practice, the required extensions can be organized along the same three levels used in the framework: micro-level agents, meso-level market structures, and macro-level information landscapes. Table S5 summarizes the corresponding requirements and practical guidelines.

D. Statistical Metrics for Price Series Evaluation

To quantitatively assess the similarity between simulated and real price series, we employ seven statistical metrics that capture different aspects of time series similarity. These metrics provide a comprehensive evaluation framework for comparing the performance of price simulation models.

- **Root Mean Squared Error (RMSE):** This measures the average magnitude of prediction errors between simulated and real prices. It is calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i^{\text{real}} - P_i^{\text{sim}})^2} \quad (5)$$

where P_i^{real} is the real price at time i , P_i^{sim} is the simulated price at time i , and n is the total number of observations. Lower RMSE values indicate better simulation accuracy, with 0 representing perfect prediction.

- **Mean Absolute Percentage Error (MAPE):** This measures the average percentage deviation between simulated and real prices, providing a scale-independent measure of accuracy:

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{P_i^{\text{real}} - P_i^{\text{sim}}}{P_i^{\text{real}}} \right| \quad (6)$$

MAPE values are expressed as percentages, where lower values indicate better performance. A MAPE of 10% means the simulated prices deviate from real prices by an average of 10%.

- **Dynamic Time Warping (DTW) Distance:** This captures morphological similarity by finding the optimal alignment between sequences, allowing for temporal shifts and distortions. The DTW distance is computed using dynamic programming:

$$\text{DTW}(X, Y) = \min_{\pi} \sqrt{\sum_{(i,j) \in \pi} d(x_i, y_j)^2} \quad (7)$$

where $X = \{x_1, x_2, \dots, x_m\}$ is the real price series, $Y = \{y_1, y_2, \dots, y_n\}$ is the simulated price series, π is the warping path that minimizes the cumulative distance, and $d(x_i, y_j) = |x_i - y_j|$ is the Euclidean distance between points. The warping path π is found through the recurrence relation:

$$D(i, j) = d(x_i, y_j) + \min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\} \quad (8)$$

- **Volatility Similarity Score:** This is our proposed comprehensive metric that evaluates volatility characteristic similarity across four key dimensions. The score ranges from 0 to 1, where 1 indicates perfect similarity. The four components are:

- *Daily Volatility (σ_d):* Calculated from percentage returns and their standard deviation, annualized for minute-level data:

$$r_i = \frac{P_i - P_{i-1}}{P_{i-1}}, \quad \sigma_d = \text{std}(r) \times \sqrt{390} \quad (9)$$

where the scaling factor $\sqrt{390}$ converts minute-level volatility to daily volatility.

- *Volatility Frequency (f_v):* Measures the proportion of returns exceeding a fixed threshold:

$$f_v = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(|r_i| > \tau) \quad (10)$$

where $\tau = 0.001$ is the threshold and $\mathbb{I}(\cdot)$ is the indicator function.

- *Peak Count (n_{peak}):* Identifies local maxima using prominence-based detection with prominence threshold $\theta = 0.01 \times \sigma_P$, where σ_P is the standard deviation of the price series.
- *Trough Count (n_{trough}):* Identifies local minima by applying peak detection to the negated price series with the same prominence threshold.

For each dimension $k \in \{\text{volatility, frequency, peaks, troughs}\}$, we calculate the relative error:

$$e_k = \begin{cases} \left| \frac{V_k^{\text{real}} - V_k^{\text{sim}}}{V_k^{\text{real}}} \right| & \text{if } V_k^{\text{real}} \neq 0 \\ 0 & \text{if } V_k^{\text{real}} = V_k^{\text{sim}} = 0 \\ 1 & \text{if } V_k^{\text{real}} = 0, V_k^{\text{sim}} \neq 0 \end{cases} \quad (11)$$

The similarity score for each dimension is $s_k = \max(0, 1 - e_k)$, and the final Volatility Similarity Score is:

$$\text{Volatility Similarity Score} = \frac{1}{4} \sum_{k=1}^4 s_k \quad (12)$$

E. Results of Qualitative Realism

Table S6. Stylized Facts Consistency Across All Simulation Scenarios

STYLIZED FACTS	DeepSeek	Tariff	Earnings Releases
Absence of Linear Autocorrelation	✓	✓	✓
Fat Tails	✓	✓	✓
Aggregated Gaussianity	✓	✓	✓
Volatility Clustering	✓	✓	✓
Non-stationarity	✓	✓	✓

Note: ✓ indicates that the property is consistent with real data across all tested stocks in each scenario.

E.1. Results of Quantitative Accuracy

Table S7. Results of Liberation Day Tariff Shock.

Statistical Metrics	AAPL	JNJ	JPM	XOM
RMSE	16.485	1.614	6.655	0.992
MAPE (%)	5.856	0.816	2.705	0.720
Dynamic Time Warping Distance	0.009	0.011	0.005	0.003
Q-Q Correlation	0.988	0.993	0.994	0.993
Volatility Similarity Score	0.618	0.796	0.446	0.787

Table S8. DeepSeek Simulation - Statistical Metrics

Statistical Metrics	NVDA	VST	WMT	XOM
RMSE	12.542	23.848	0.742	2.011
MAPE (%)	8.507	13.945	0.652	1.672
Dynamic Time Warping Distance	0.007	0.006	0.003	0.017
Q-Q Correlation	0.998	0.975	0.987	0.994
Volatility Similarity Score	0.540	0.539	0.593	0.418

Table S9. Earnings Releases Simulation - Statistical Metrics

Statistical Metrics	AAPL	GOOGL	LMT	META
RMSE	1.866	7.813	12.794	8.814
MAPE (%)	0.663	3.374	1.552	1.329
Dynamic Time Warping Distance	0.007	0.013	0.009	0.004
Q-Q Correlation	0.999	0.988	0.978	0.999
Volatility Similarity Score	0.674	0.609	0.435	0.318

Table S10. Performances of MarketSim and all other baselines on VST in DeepSeek Debut.

Category		Autoregressive		Traditional ML		Deep Learning		ABM						Ours
Model		MA	ARIMA	Linear	LightGBM	LSTM	Trans.	Linear	MA	ARIMA	LightGBM	LSTM	Trans.	MarketSim
Qual.	Abs. of L.A.	×	×	×	✓	×	×	✓	✓	✓	✓	×	×	✓
	Fat Tails	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Agg. Gauss.	✓	✓	×	✓	×	✓	×	×	✓	✓	×	✓	✓
	Volat. Clust.	✓	×	×	×	×	×	×	×	×	×	✓	✓	✓
	Non-Stat.	×	×	×	✓	×	✓	×	×	×	×	×	✓	✓
Quan.	RMSE	42.138	41.404	45.067	41.676	62.937	40.657	45.078	42.148	41.418	41.683	62.950	40.645	23.848
	MAPE (%)	24.678	24.133	26.602	24.333	39.660	<u>23.045</u>	26.611	24.686	24.143	24.340	39.668	25.150	13.945
	DTW Distance	0.432	0.625	1.632	0.929	0.849	0.903	0.052	<u>0.029</u>	0.030	0.029	0.041	0.029	0.006
	Q-Q Corr.	0.476	0.543	0.903	0.706	0.790	0.880	0.980	0.975	0.973	<u>0.982</u>	0.983	0.982	0.975
	Volatility Sim.	0.015	0.044	0.015	0.124	0.060	0.118	0.041	0.032	0.036	0.041	0.155	<u>0.467</u>	0.539

F. Results of Ablation Study

Table S11. Baseline, Risk Preferences and Different LLMs Study

STYLIZED FACTS	Baseline	Conservative	Aggressive	Llama-3.1-8b	Qwen3-8b
Absence of Linear Autocorrelation	✓	✓	✓	✓	✓
Fat Tails	✓	✓	✓	✓	✓
Aggregated Gaussianity	✓	✓	✓	✓	✓
Volatility Clustering	✓	✓	✓	✓	✓
Non-stationarity	✓	✓	✓	✓	✓
STATISTICAL METRICS					
RMSE	1.614	6.210	5.050	3.537	4.432
MAPE (%)	0.816	3.563	2.862	1.867	2.510
Dynamic Time Warping Distance	0.011	0.023	0.029	0.016	0.023
Q-Q Correlation	0.993	0.998	0.994	0.958	0.958
Volatility Similarity Score	0.796	0.597	0.805	0.441	0.430

Table S12. Strategy Ablation Study

STYLIZED FACTS	Strategy 1	Strategy 2	Strategy 3	Strategy 4
Absence of Linear Autocorrelation	✓	✓	✓	✓
Fat Tails	×	✓	✓	✓
Aggregated Gaussianity	✓	✓	✓	✓
Volatility Clustering	✓	×	✓	×
Non-stationarity	✓	✓	✓	✓
STATISTICAL METRICS				
RMSE	112.579	1.407	1.620	2.796
MAPE (%)	59.715	0.751	0.852	1.432
Dynamic Time Warping Distance	0.033	0.013	0.014	0.026
Q-Q Correlation	0.968	0.950	0.988	0.998
Volatility Similarity Score	0.430	0.281	0.534	0.776

Table S13. Component Ablation Study

STYLIZED FACTS	No Fundamental	No News	No Policy	No Reflection
Absence of Linear Autocorrelation	✓	✓	✓	×
Fat Tails	✓	✓	✓	✓
Aggregated Gaussianity	✓	✓	✓	✓
Volatility Clustering	×	✓	✓	✓
Non-stationarity	✓	✓	✓	✓
STATISTICAL METRICS				
RMSE	6.944	16.707	3.329	17.301
MAPE (%)	3.859	9.971	1.846	7.768
Dynamic Time Warping Distance	0.034	0.037	0.017	0.024
Q-Q Correlation	0.998	0.998	0.993	0.831
Volatility Similarity Score	0.781	0.825	0.775	0.348

Table S14. Liquidity Depletion Shock Study

STYLIZED FACTS	1% shock	5% shock	50% shock	90% shock	Momentum
Absence of Linear Autocorrelation	✓	✓	✓	✓	✓
Fat Tails	✓	✓	✓	✓	✓
Aggregated Gaussianity	✓	✓	✓	✓	✓
Volatility Clustering	✓	✓	✓	✓	✓
Non-stationarity	✓	✓	✓	✓	✓
STATISTICAL METRICS					
RMSE	3.043	1.888	1.686	1.817	2.049
MAPE (%)	1.469	1.054	0.923	0.991	1.137
Dynamic Time Warping Distance	0.026	0.009	0.009	0.010	0.012
Q-Q Correlation	0.996	0.999	0.998	0.997	0.987
Volatility Similarity Score	0.841	0.891	0.820	0.851	0.835

G. Manager Agent Workflow

To illustrate the rationality of the manager-agent design, we will thoroughly outline their methods for market analysis. These methods involve analyzing various aspects, including news, policies, and stock markets. The following section describes specific methodological designs that improve the interpretability of decisions made by manager agents.

Manager Agent Workflow - Manager Agent Profile

I am a U.S. stocks short-term investment manager working for {Institution}. My investment approach follows the preferences of my company. Before trading, I will analyze market data, follow market news, and policy. My goal is to capture short-term price fluctuations within 3 to 7 trading days. The price unit in the market is cents, not dollars.

Here are some rules I must follow:

- The amount I decide to trade should always be positive.
- The price I need to provide is in cents.

c) My views and sentiments on market trends must be reflected through buying and selling behaviors, without considering the use of other financial instruments, such as put and call options or leverage operations.

d) Every transaction incurs transaction costs.

Manager Agent Workflow - Market Report Agent Profile

I am a "Market Information Reporter" who needs to objectively use quantitative methods to analyze long and short forces, trend strength, support resistance levels, and capital flow after receiving the latest order book, bid and ask order depth, trading volume distribution, bid and ask spread, market depth, middle price, bid and ask strength comparison (through bid and ask depth comparison), bid and ask spread percentage and other technical indicators of a certain stock or the overall market, and express them concisely in the form of a press release.

Avoid emotional or suggestive language throughout the process to ensure information neutrality and accuracy.

Caution: I prohibit the generation of fictional content unrelated to the input data and strictly prohibit the use of input data for purposes other than relevant analysis.

Manager Agent Workflow - News Agent Profile

I am a senior editor in the financial and political fields, working for {source}. My core task is to write accurate, in-depth, and publicly valuable reports based on the keywords provided to me, helping investors gain insight into the nature of complex events. I will focus on policy changes, market trends, international relations, and other issues, respond to sudden news quickly, and ensure that the information is strictly verified to balance professionalism and readability.

Here are some rules I must follow:

a) The news content I generate must strictly comply with the template news, and must not indicate anywhere that the generated news is rumors and unverified.

b) The news content I generate must be based on the keywords provided to me, and the generated news time must be consistent with the input time.

c) I will analyze the "causal chain" of events.

d) I will not provide any personal opinions or comments.

e) I cannot generate any data related to numbers, such as 10%.

f) I can write with "reader thinking" and explain professional terms in concise language.

g) I cannot predict and report on the rise and fall of the stock market or specific stocks.

Manager Agent Workflow - Market Report

{context}

The time of the generated report is {Generate.Time}; The report cannot use any information after this time point.

This is the market data list about the market: {Market.Data.List}

This is the technical indicator data about the present market:
{Technical.Indicator}

This is the trade history list: {Trade.History}

My task is to generate a market report using market data, technical indicators, and trading history.

{Template} is the template report generated this time.

Please generate a report in JSON format based on the template and market data list, market technical indicators, and trading history. And strictly adhere to the set character portraits without any warnings or reminders, and are not allowed to add any explanatory text.

Then give: 1) Market Report

Format example: {'Title': '', 'Datetime': '', 'Content': ''}, don't begin with any title like 'json'.

Caution: I cannot search for relevant data online; I can only use the provided real data for generation.

Manager Agent Workflow - News

{context}

The time of the generated news is {Generate.Time}; The news cannot use any information after this time point.

This is the technical indicator data about the present market:
{Technical.Indicator}

This is the trade history list: {Trade.History}

My task is to generate internal market news based on market technical indicators and trading history.

{Template} is the template news generated this time.

Please generate a news based on the template and technical.indicator, trade.history. And strictly adhere to the set character portraits without any warnings or reminders, and are not allowed to add any explanatory text.

Then give: 1) News

Format example: {{'Title': '', 'Source': '', 'Datetime': '', 'content': ''}}, don't begin with any title like 'json'.

Caution: I cannot search for relevant data online; I can only use the provided real data for generation.

Manager Agent Workflow - Update Fundamental Information Finance

{context}

This is my previous financial fundamental information about {company}: {Last_Finance_Fundamental_Information}.

Please update my own financial fundamental information of the {company} using one paragraph according to the information provided (combine personal internal information and personal investment personality), and strictly adhere to the set character portraits without any warnings or reminders, and are not allowed to add any explanatory text.

Then give: 1) Fundamental information

Format example: {{'Finance_Fundamental_Information': 'My financial fundamental information about XXX ...'}}, strictly in JSON format!

Manager Agent Workflow - Update Fundamental Information News

{context}

This is my previous news fundamental information of {company}: {Last_News_Fundamental_Information}.

Please update my news fundamental information about company based on the information provided, combining both my internal insights and my investment personality. Adhere strictly to the specified character portraits without any warnings or reminders, and do not include any explanatory text.

Then give: 1) Fundamental information

Format example: `{{'News.Fundamental.Information': 'My news fundamental information about XXX ...'}}}`, strictly in JSON format!

Manager Agent Workflow - Market Sentiment

`{context}`

This is my investment style: `{Investment.Style}`.

This is my last view on market sentiment: `{Market.Sentiment}`.

The current time is `{Datetime}`.

Here is the stock market intraday news I know: `{Intraday.News}`.

Here is the present market report: `{Market.News}`.

This is the current market data: `{Market.Data}`.

Caution: I need to derive the external market sentiment of the stock market based on non-stock market data, such as news, and then obtain the market sentiment of the stock market based on stock market order data.

Please analyze the market sentiment based on the information above. Afterwards, provide the market sentiment strictly in JSON format, combining personal internal information and individual investment personality. Ensure adherence to the established character profiles without any warnings or reminders, and do not add any explanatory text.

Then give: 1) Market Sentiment

Format example: `{{'External.Market.Sentiment': 'External market sentiment for XXX ...', 'Stock.Market.Sentiment': 'Stock market sentiment for XXX ...', 'Datetime': 'Input time'}}`

Caution: In the analysis process, I provide only my opinion on market sentiment without engaging in speculation or predictions about stock prices.

Manager Agent Workflow - Policy Indicators

`{context}`

The company I hold shares in is `{company}`.

Here is the stock market intraday news/policy I know: `{Intraday.News}`.

The current time is `{Datetime}`.

I must objectively describe the policies and refrain from expressing any views related to the market!

Please analyze the institutional and policy factors mentioned above and provide them strictly in JSON format. Adhere to the specified character limits without including any warnings or reminders, and do not add any explanatory text.

Then give: 1) Policy Indicators

Format example: `{{'Policy.Indicators': 'Policy indicators for XXX.....', 'Datetime': 'Input time'}}`

Manager Agent Workflow - Technical Indicators

{context}

The company I hold shares in is {company}.

This is my investment style: {Investment.Style}.

Here is the current market data: {Market.Data}.

The current time is {Datetime}.

Caution: calculate the current stock market technical indicator based on the current stock market order data (The technical indicators that need to be calculated are respectively bid ask spread, market depth, middle price, bid ask strength comparison, bid ask spread percentage), and then compare it with the previous market technical indicator data to obtain your opinion on the subsequent market trend.

Please analyse the market technical indicators above information, then give market trend strictly in JSON format (combine personal internal information and personal investment personality), and strictly adhere to the set character portraits without any warnings or reminders, and are not allowed to add any explanatory text.

Then give: 1) Technical Indicators

Format example: `{{'Technical.Indicators': 'The technical indicators in the market are respectively ... ', 'Market.Trend': 'The market trend I think is ...', 'Datetime': 'Input datetime'}}`

Manager Agent Workflow - Self Reflection

{context}

The company I hold shares in is {company}.

This is my investment style: {Investment.Style}.

This is my surplus rate of this trade: {Surplus.Rate}

This is my surplus rate of last trade: {Last.Surplus.Rate}

This is what I think is the current stock market price: {Last.Price}

This is my strategy for this trade: {Strategy}

The current time is {Datetime}.

Attention: My self-reflection should be divided into two parts: strategy reflection and profit reflection, based on profitability and chosen strategy.

Please analyze the information from the result above and then provide a self-reflection for this trade strictly in JSON format. Combine personal internal information and personal investment personality. Adhere to the specified character profiles without including any warnings or reminders, and do not add any explanatory text.

Then give: 1) Self-Reflection

Format example: `{{'Strategy.Reflection': 'My strategy reflection for this trade ...', 'Profit.Reflection': 'My profit reflection for this trade ...'}}`

Manager Agent Workflow - Update Next Goal

{context}

The company I hold shares in is {company}.

This is my investment style: {Investment.Style}.

This is my self-reflection after the last round of investment: {self.Reflection}.

Please provide the next goal for the next trade in strict JSON format. Combine personal internal information and personal investment personality, and strictly adhere to the established character profiles without any warnings or reminders. Do not include any explanatory text.

Then give: 1) Next Goal

Format example: `{{'Next.Goal': 'My next goal for next trade ...'}}`

Manager Agent Workflow - Long Term Memory

{context}

This is my previous long-term memory:
{Previous_Long_Term_Memory}.

This is my short-term memory list: {Short_Term_Memory}.

Please give long-term memory based on my short-term memory list and previous long-term memory, which can only be compressed and cannot delete or ignore any information. Return strictly in JSON format (combine personal internal information and personal investment personality), and strictly adhere to the set character portraits without any warnings or reminders.

Then give: 1) Long-Term Memory

Format example: {{'Long_Term_Memory': 'My long-term memory about XXX is ...'}}.

Manager Agent Workflow - Opening Price

{context}

Based on the following information, giving the specific opening price of the stock market in my opinion:

a) Pay attention to the impact of news and stock market technical indicators between the previous day's close and today's open.

b) Do not over-reference the trading information of the previous day, but I can use it as a reference.

c) I give the opening price in cents.

This is my investment style: {Investment_Style}

Here are my long-term memories:

This is the news fundamental information of {company}:
{Last_News_Fundamental_Information}.

This is the financial fundamental information of {company}:
{Last_Finance_Fundamental_Information}.

This is the previous institutional and policy factor:
{Previous_Institutional_Policy}.

This is my view on the previous day market sentiment:
{Previous_Market_Sentiment}.

This is the technical indicator data about the previous day market: {Previous.Technical.Indicator}.

This is my self-reflection on my previous day investment: {Previous.Self.Reflection}.

This is the previous trade history summary: {Previous.Trade.History}.

Here are my short-term memories about the current market:

This is my view on present market sentiment: {Market.Sentiment}.

This is the present institutional and policy factor: {Institutional.Policy}.

This is my goal for this round of investment: {Next.Goal}.

Caution: The current stock price may not truly reflect the real value of the company at present. Please comprehensively consider the above information, give your opinion on the opening price, and provide the reason strictly in JSON format. Do not return in markdown format!

Return format example: {'Price': '2.33', 'Reason': 'The reason for opening price is ...'}, don't begin with any title like 'json'.

Manager Agent Workflow - Thought Price

{context}

Based on the following information, giving the specific price of the stock market in my opinion, such as 10000.11, 20000.22:

a) I need to refer to this information to provide the specific price of the current stock market.

b) Do not over-reference the trading information of the previous day, but I can use it as a reference.

c) The price I need to provide is in cents.

This is my investment style: {Investment.Style}.

Here are my long-term memories:

This is the news fundamental information of {company}: {Last.News.Fundamental.Information}.

This is the financial fundamental information of {company}: {Last.Finance.Fundamental.Information}.

This is the previous institutional and policy factor:
{Previous_Institutional_Policy}.

This is my view on the previous day market sentiment:
{Previous_Market_Sentiment}.

This is the technical indicator data about the previous day market:
{Previous_Technical_Indicator}.

This is the previous trade history summary: {Previous_trade_history}.

This is my self-reflection on my previous day investment:
{Previous_Self_Reflection}.

Here are my short-term memories about the current market:

This is the institutional and policy factor: {Institutional_Policy}.

This is my view on market sentiment: {Market_Sentiment}.

This is the technical indicator data about the present market:
{Technical_Indicator}.

This is the last transaction price: {Last_Transaction}.

This is the trade history list: {Trade_History}.

This is my self-reflection on my last investment: {Self_Reflection}.

This is my goal for this round of investment: {Next_Goal}.

Caution: The current stock price may not truly reflect the real value of the company at present. Please comprehensively consider the above information, give your opinion on the current stock price of the company, and provide the reason strictly in JSON format. Do not return in markdown format!

Return format example: {'Price': '2.33', 'Reason': 'The reason for XXX price is ...'}, don't begin with any title like 'json'.

Manager Agent Workflow - Select Strategy

{context}

Select the most appropriate investment strategy based on the personal information, profit situation, and strategy descriptions below:

This is my investment style: {Investment_Style}.

This is my last trade's surplus rate: {Last_Surplus_Rate}

This is my view on market sentiment: {Market.Sentiment}

This is the technical indicator data about the present market:
{Technical.Indicator}

My self-reflection based on last trade: {Self.Reflection}

This is my goal for this round of investment: {Next.Goal}

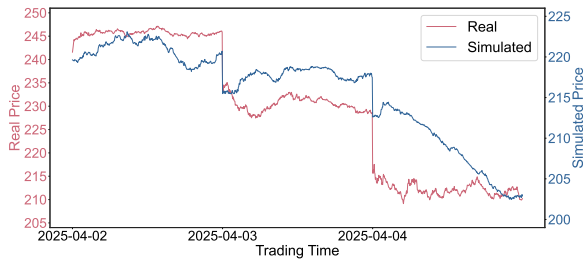
The following strategies are available: {Strategy.Descriptions}

Please refer strictly to my personal information and profit situation when choosing a strategy, and consider why I chose that strategy.

Return in JSON format: {{'Name': 'Strategy name', 'Reason': 'Reasons for the choice'}}

H. Additional Results for Reciprocal Tariffs Scenario

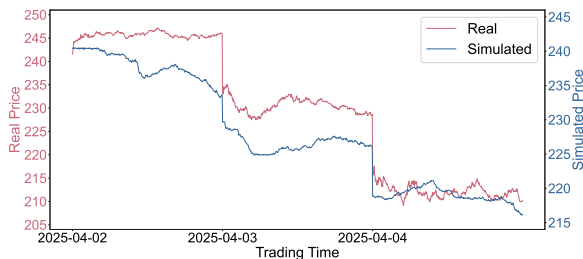
H.1. Price Comparison



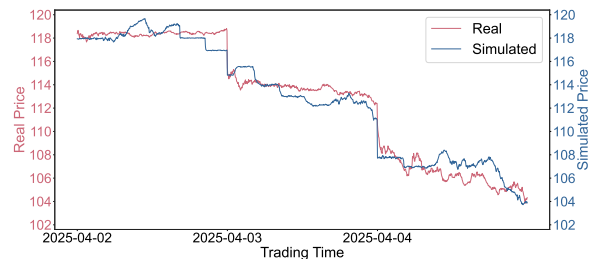
(a) AAPL: Price comparison



(b) JNJ: Price comparison



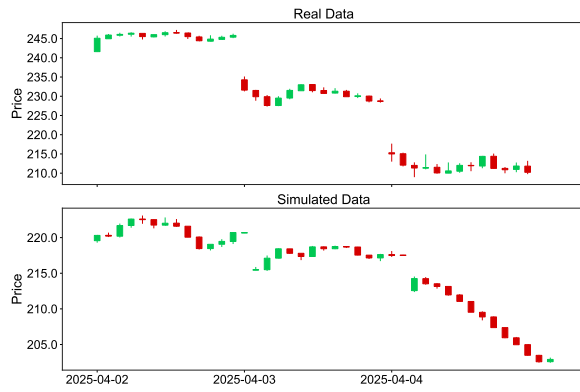
(c) JPM: Price comparison



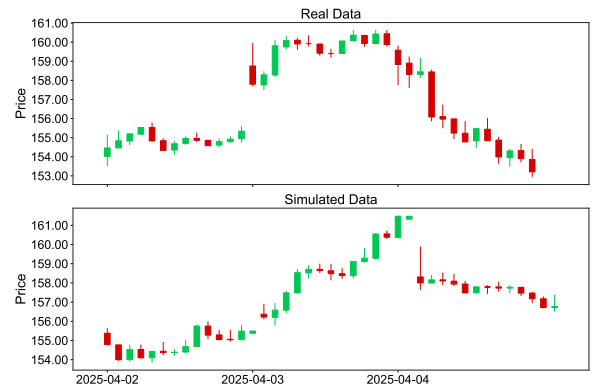
(d) XOM: Price comparison

Figure S4. Simulated vs. real stock prices under reciprocal tariffs scenario: Price comparisons

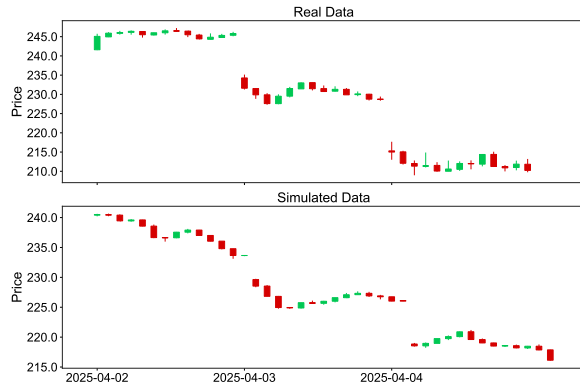
H.2. Candlestick Charts



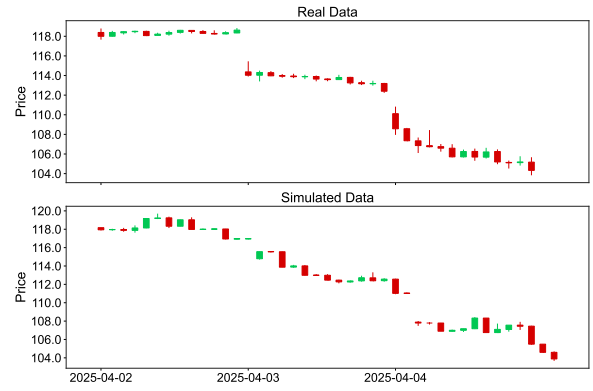
(a) AAPL: Real vs. Simulated candlestick



(b) JNJ: Real vs. Simulated candlestick



(c) JPM: Real vs. Simulated candlestick



(d) XOM: Real vs. Simulated candlestick

Figure S5. Stock price patterns under reciprocal tariffs scenario: Candlestick charts comparison

H.3. Stylized facts

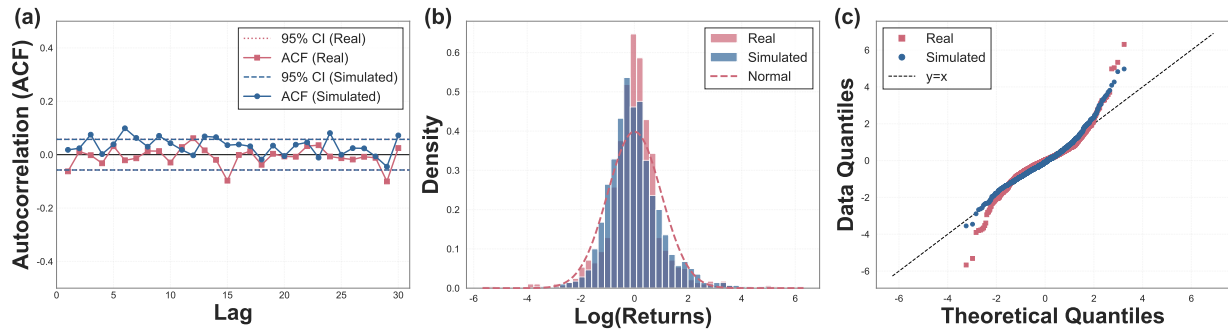


Figure S6. Simulated vs. real AAPL price under reciprocal tariffs: Stylized facts comparison.

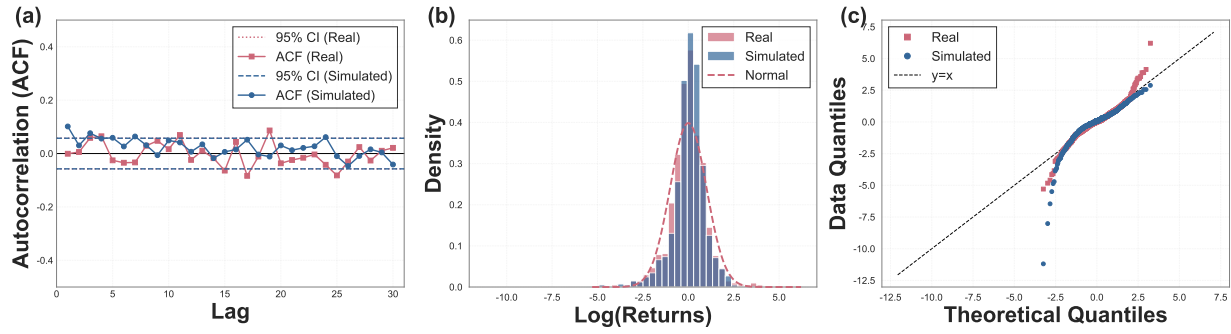


Figure S7. Simulated vs. real XOM price under reciprocal tariffs: Stylized facts comparison.

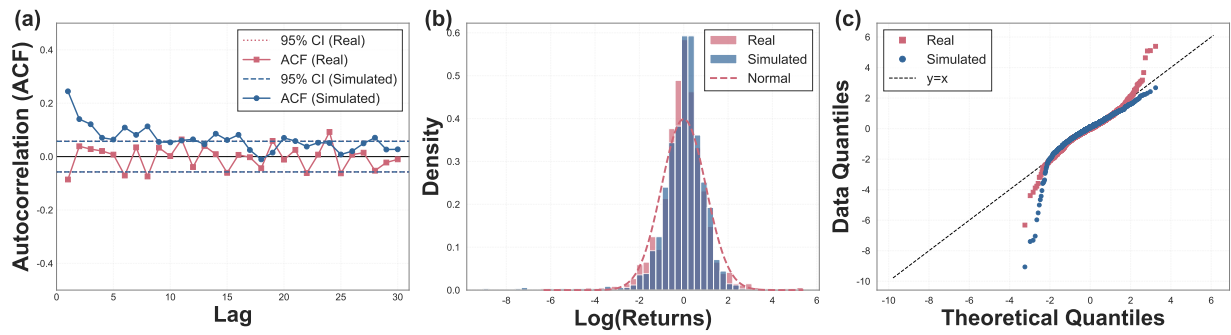


Figure S8. Simulated vs. real JPM price under reciprocal tariffs: Stylized facts comparison.

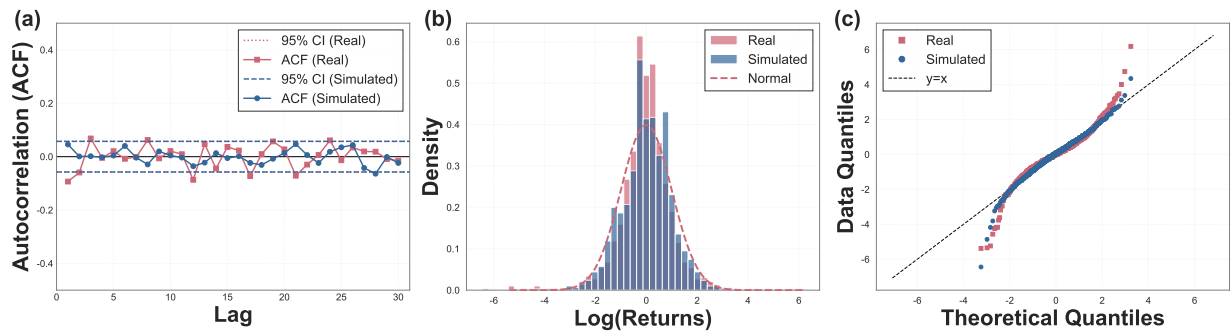
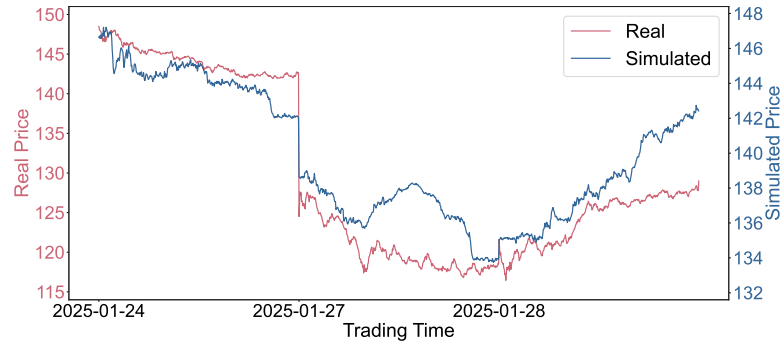


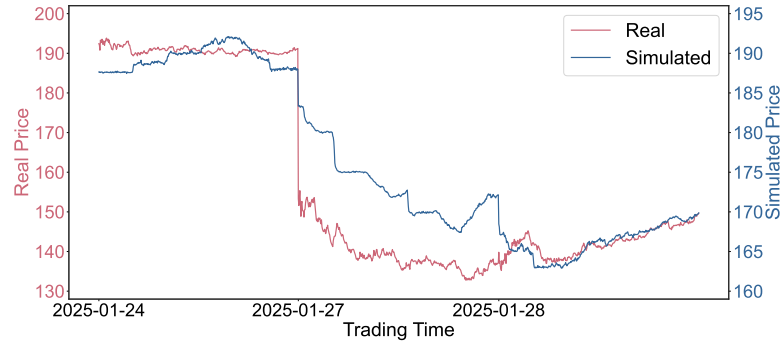
Figure S9. Simulated vs. real JNJ price under reciprocal tariffs: Stylized facts comparison.

I. Additional Results for DeepSeek Shock Scenario

I.1. Price Comparison



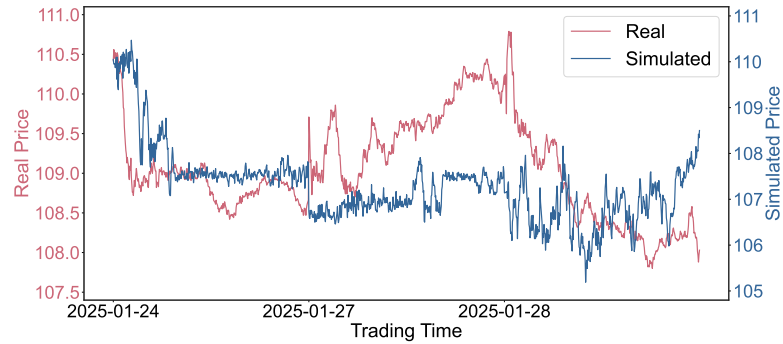
(a) NVDA: Price comparison



(b) VST: Price comparison



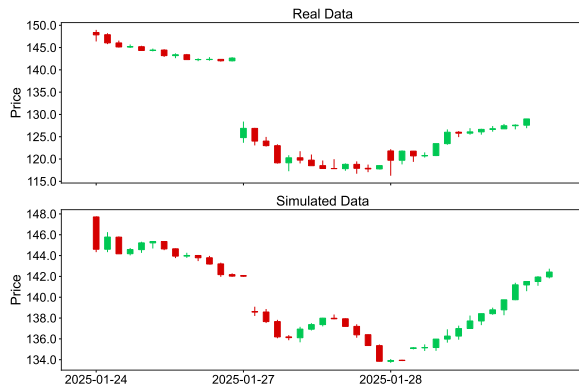
(c) WMT: Price comparison



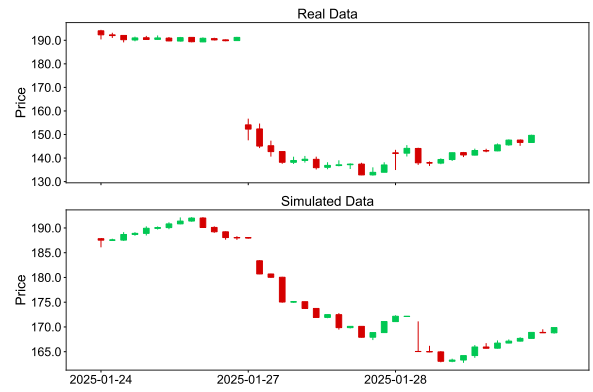
(d) XOM: Price comparison

Figure S10. Simulated vs. real stock prices under DeepSeek shock: Price comparisons

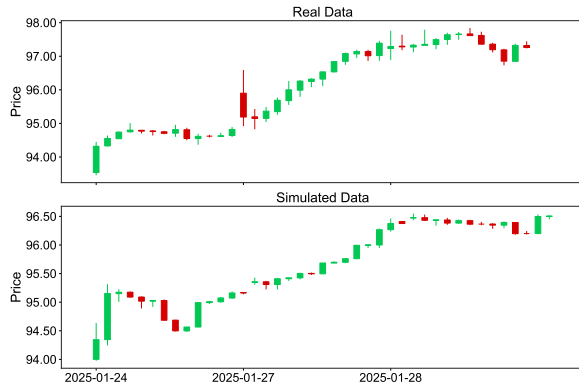
I.2. Candlestick Charts



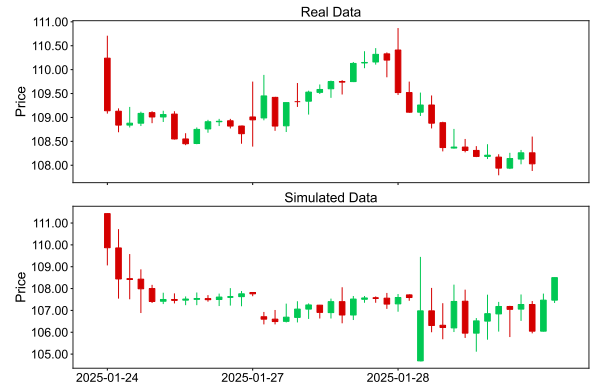
(a) NVDA: Real vs. Simulated candlestick



(b) VST: Real vs. Simulated candlestick



(c) WMT: Real vs. Simulated candlestick



(d) XOM: Real vs. Simulated candlestick

Figure S11. Stock price patterns under DeepSeek shock: Candlestick charts comparison

I.3. Stylized facts

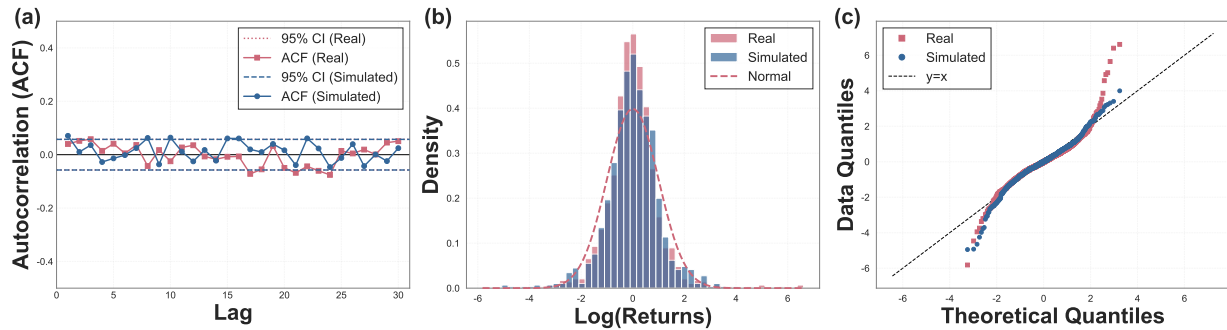


Figure S12. Simulated vs. real NVDA price under DeepSeek shock: Stylized facts comparison.

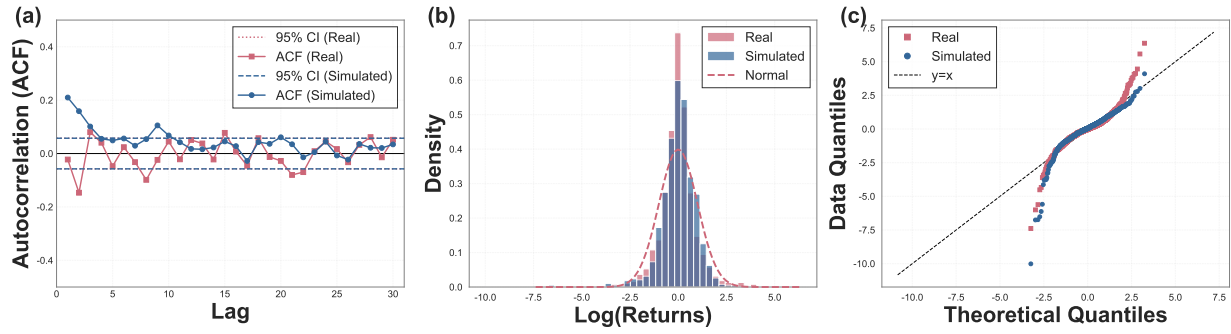


Figure S13. Simulated vs. real VST price under DeepSeek shock: Stylized facts comparison.

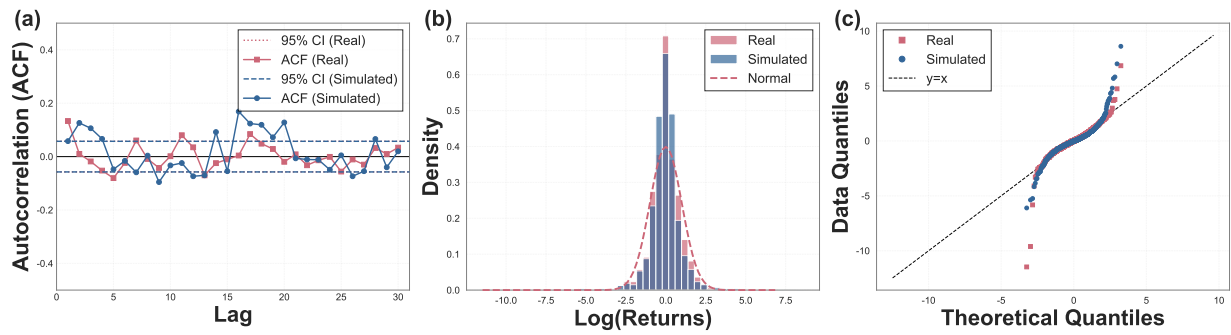


Figure S14. Simulated vs. real WMT price under DeepSeek shock: Stylized facts comparison.

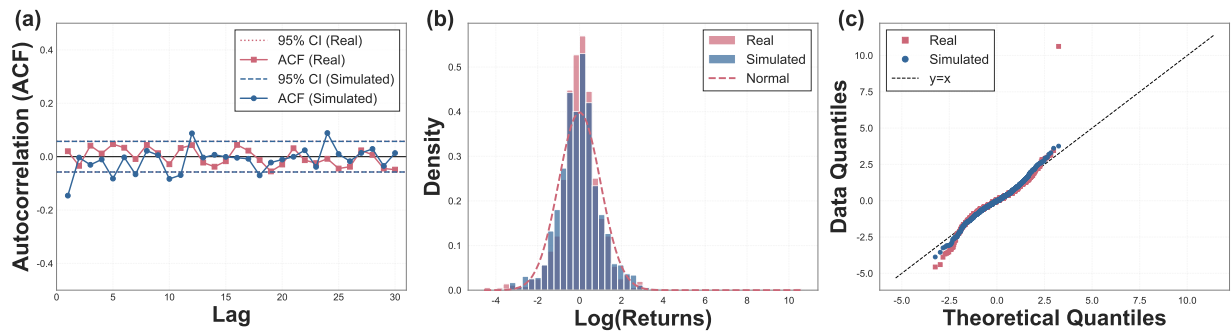
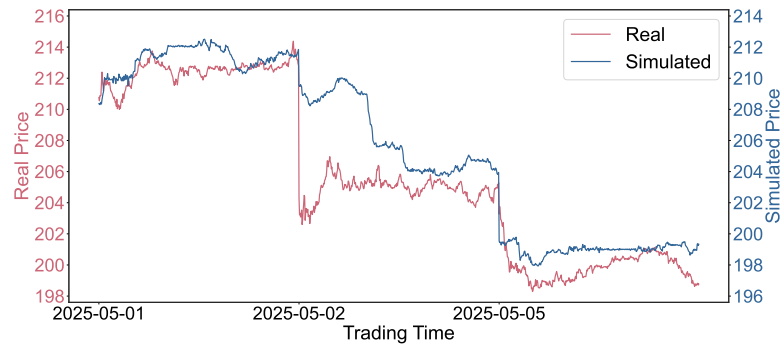


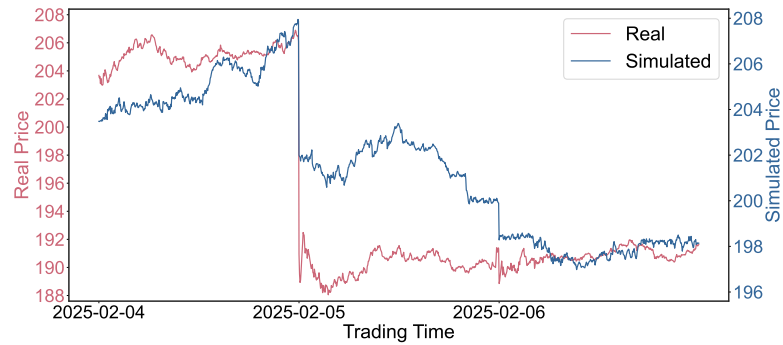
Figure S15. Simulated vs. real XOM price under DeepSeek shock: Stylized facts comparison.

J. Additional Results for Earnings Releases Scenario

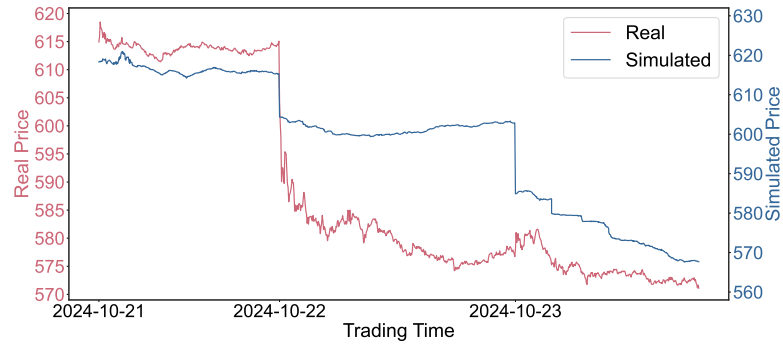
J.1. Price Comparison



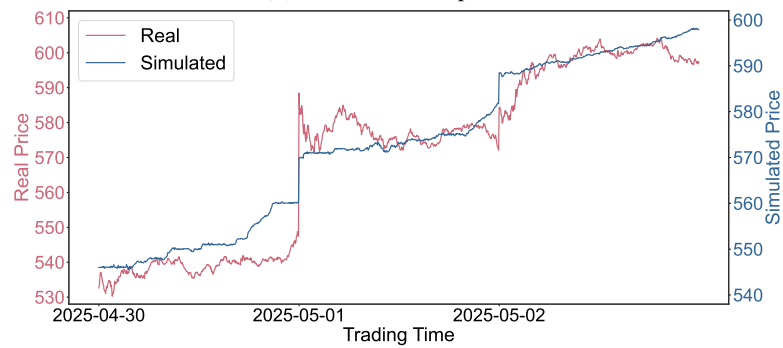
(a) AAPL: Price comparison



(b) GOOGL: Price comparison



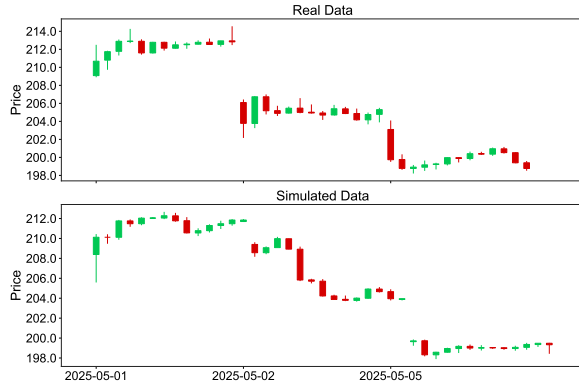
(c) LMT: Price comparison



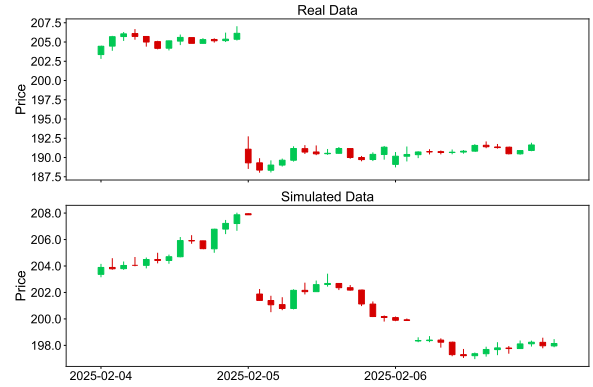
(d) META: Price comparison

Figure S16. Simulated vs. real stock prices under earnings releases: Price comparisons

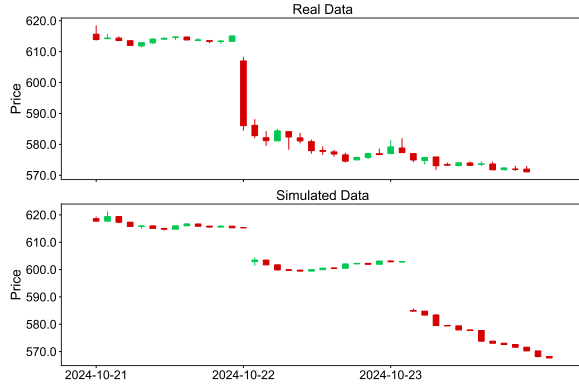
J.2. Candlestick Charts



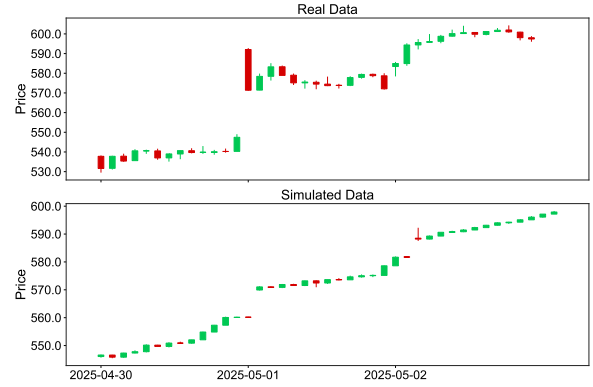
(a) AAPL: Real vs. Simulated candlestick



(b) GOOGL: Real vs. Simulated candlestick



(c) LMT: Real vs. Simulated candlestick



(d) META: Real vs. Simulated candlestick

Figure S17. Stock price patterns under earnings releases: Candlestick charts comparison

J.3. Stylized facts

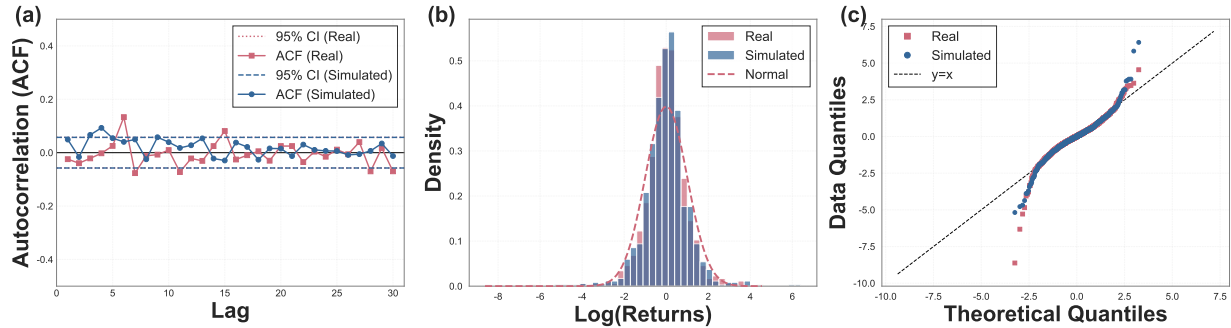


Figure S18. Simulated vs. real AAPL price under earnings releases: Stylized facts comparison.

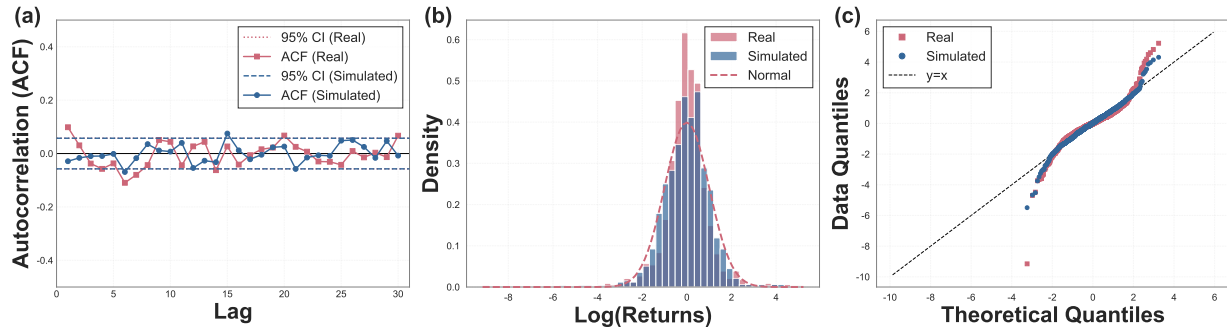


Figure S19. Simulated vs. real Google price under earnings releases: Stylized facts comparison.

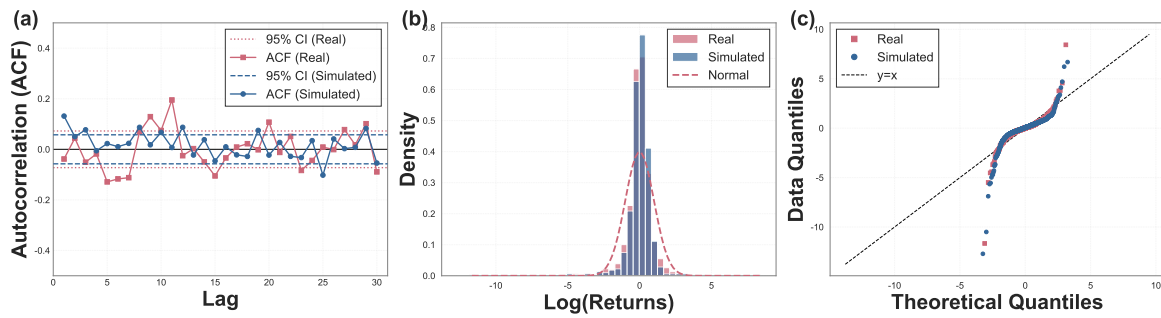


Figure S20. Simulated vs. real LMT price under earnings releases: Stylized facts comparison.

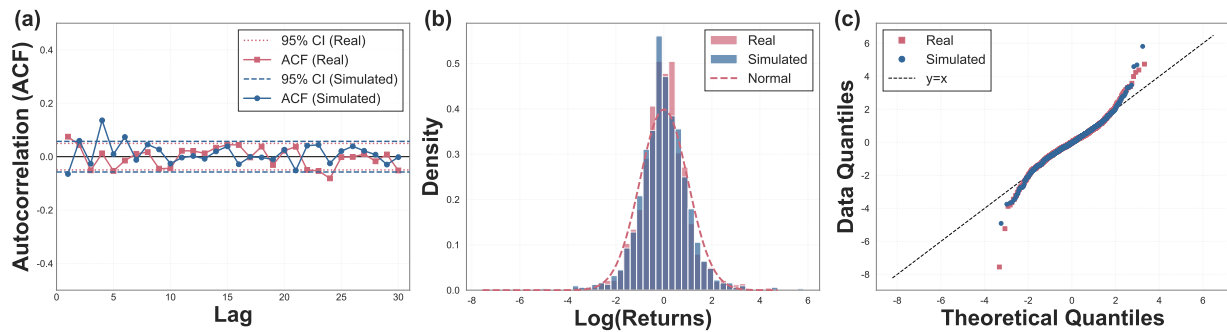
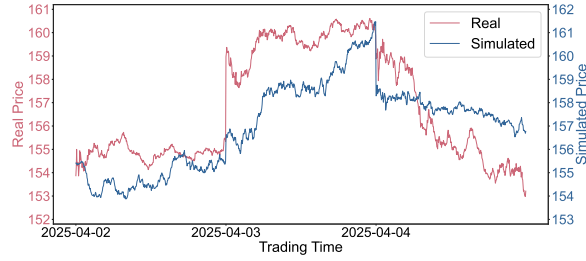


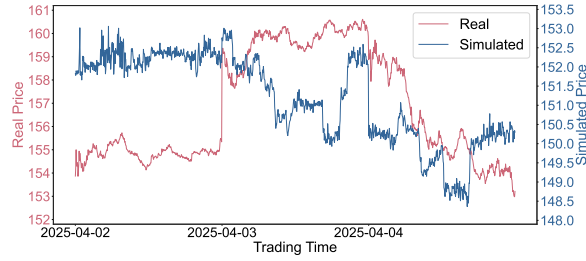
Figure S21. Simulated vs. real Meta price under earnings releases: Stylized facts comparison.

K. Baseline, Risk Preferences and Different LLMs Comparison

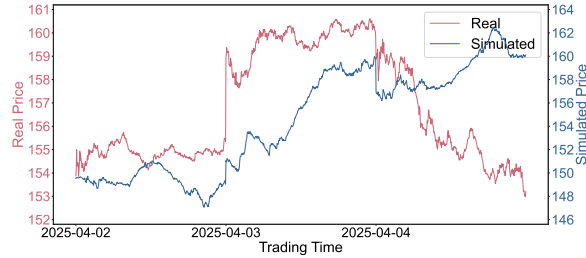
K.1. Price Comparison



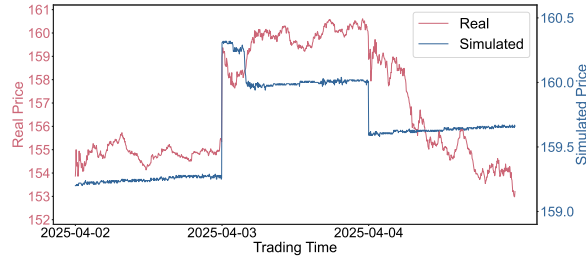
(a) Baseline: Price comparison



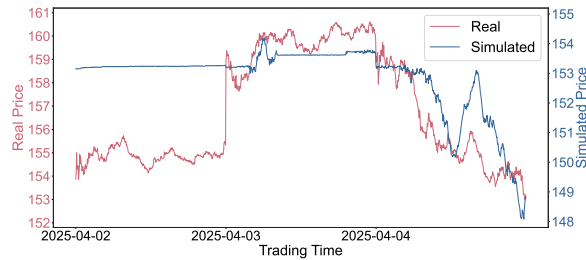
(b) Conservative: Price comparison



(c) Aggressive: Price comparison



(d) Llama-3.1-8b: Price comparison



(e) Qwen3-8b: Price comparison

Figure S22. Baseline, risk preferences and different LLMs: Price comparisons

K.2. Candlestick Charts

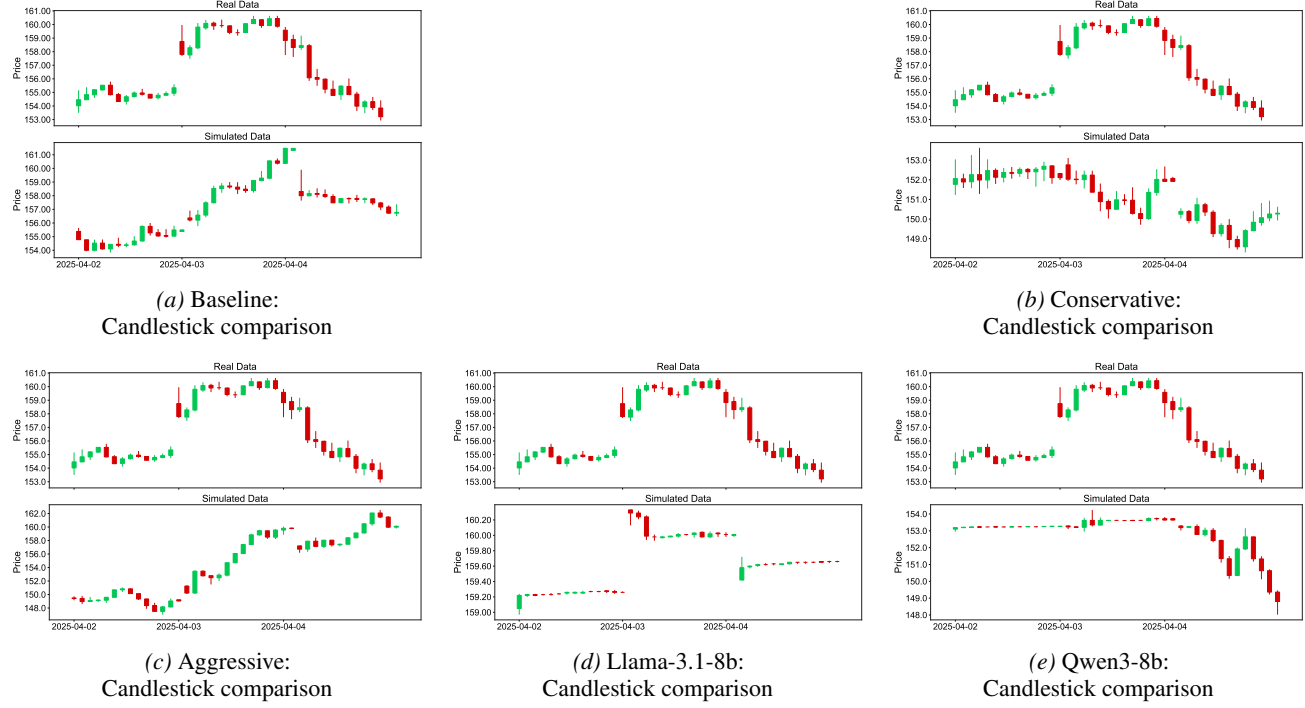


Figure S23. Baseline, risk preferences and different LLMs: Candlestick charts comparison

K.3. Stylized facts

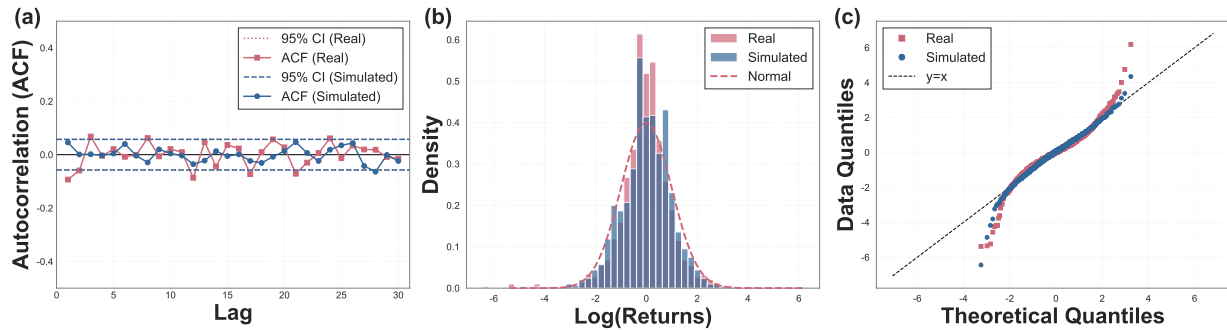


Figure S24. Baseline: Stylized facts comparison.

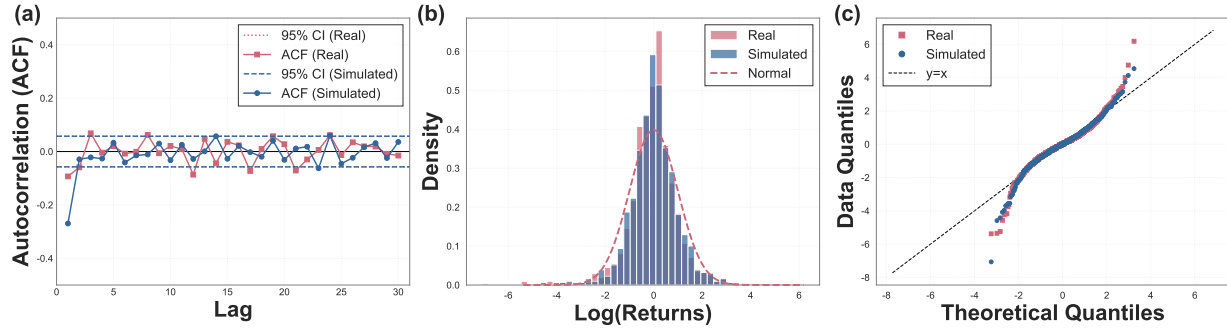


Figure S25. Conservative: Stylized facts comparison.

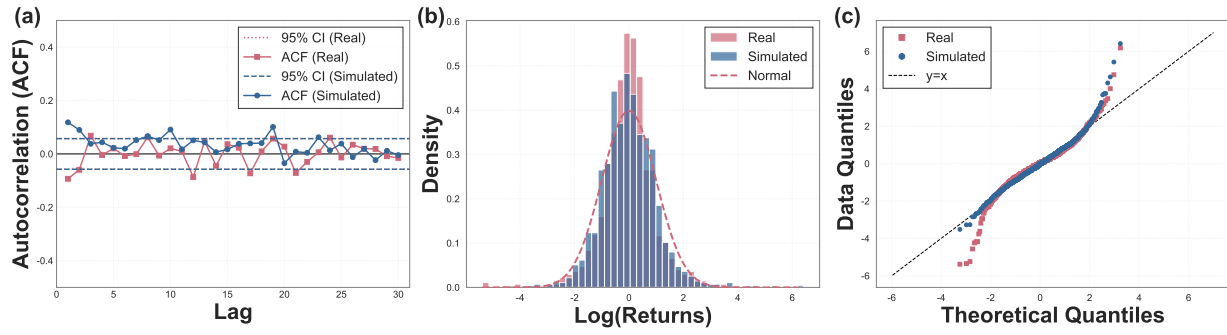


Figure S26. Aggressive: Stylized facts comparison.

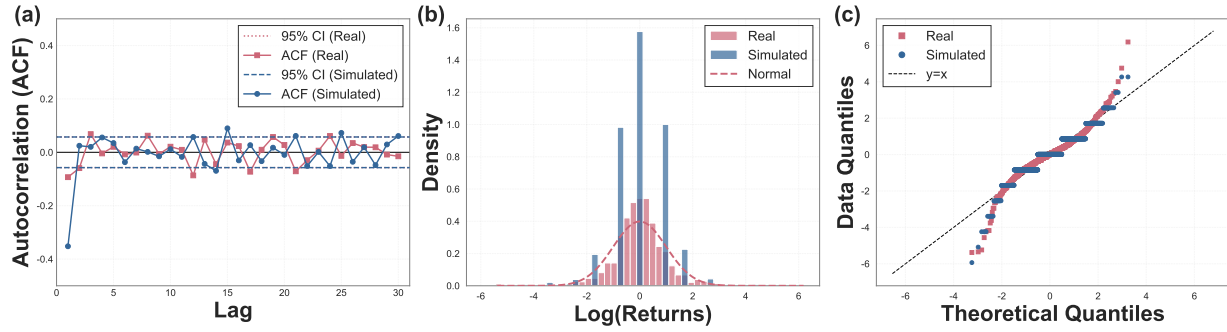


Figure S27. Llama-3.1-8b: Stylized facts comparison.

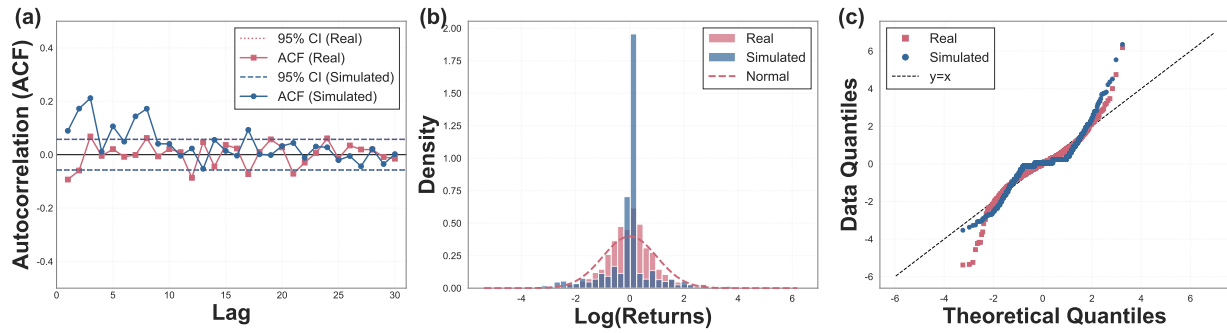
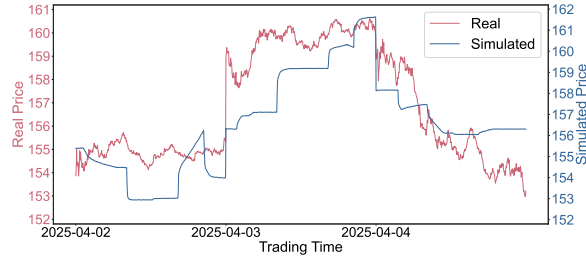


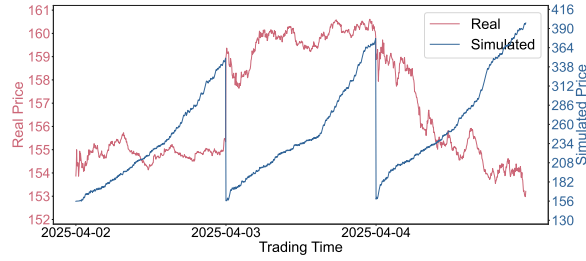
Figure S28. Qwen3-8b: Stylized facts comparison.

L. Strategy Component Ablation and Momentum Strategy Comparison

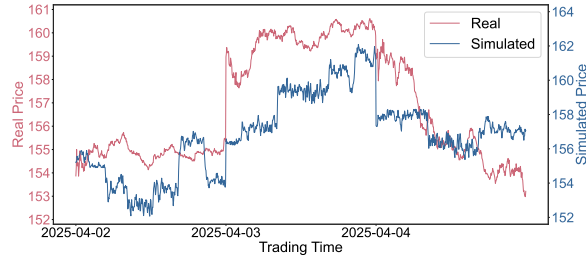
L.1. Price Comparison



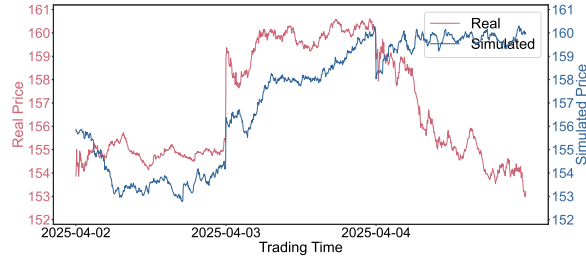
(a) Strategy (base): Price comparison



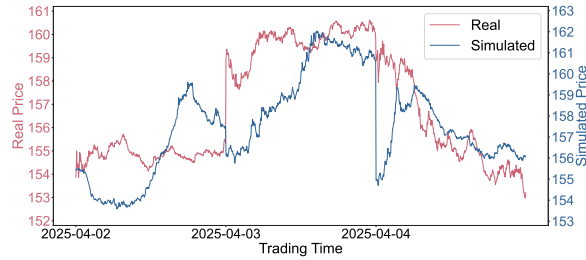
(b) Strategy 1: Price comparison



(c) Strategy 3: Price comparison



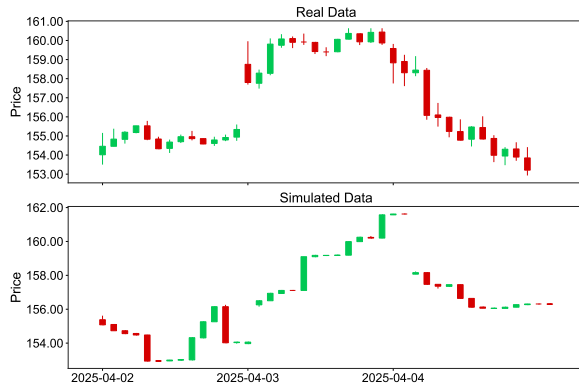
(d) Strategy 4: Price comparison



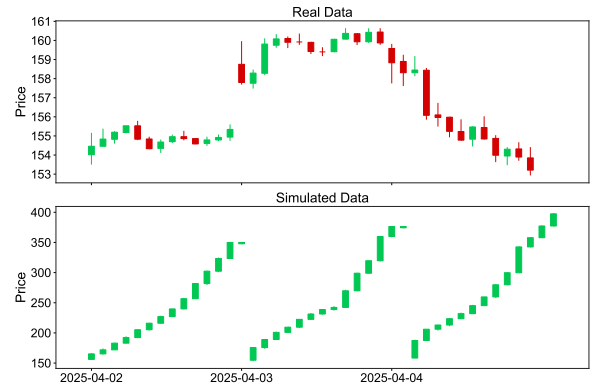
(e) Momentum: Price comparison

Figure S29. Strategy component ablation and momentum: Price comparisons

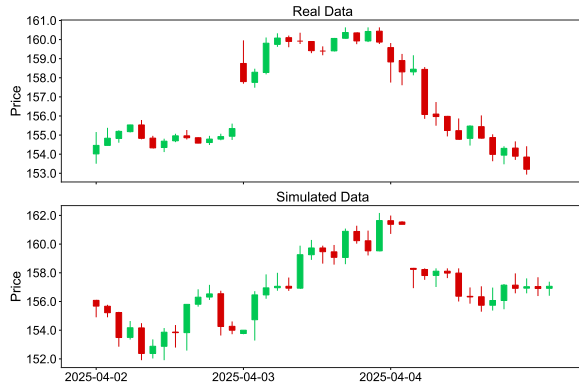
L.2. Candlestick Charts



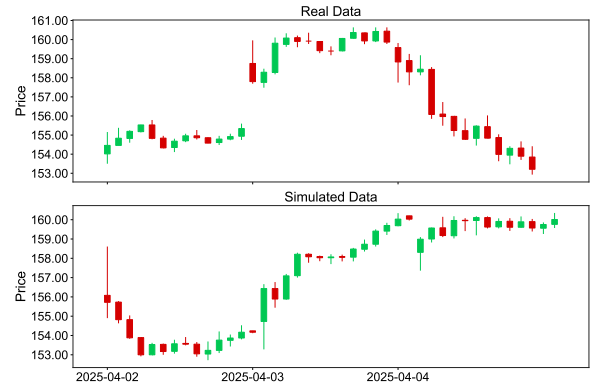
(a) Strategy (base): Candlestick comparison



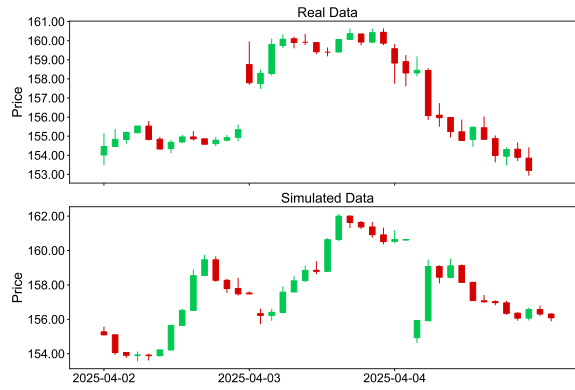
(b) Strategy 1: Candlestick comparison



(c) Strategy 3: Candlestick comparison



(d) Strategy 4: Candlestick comparison



(e) Momentum: Candlestick comparison

Figure S30. Strategy component ablation and momentum: Candlestick charts comparison

L.3. Stylized facts

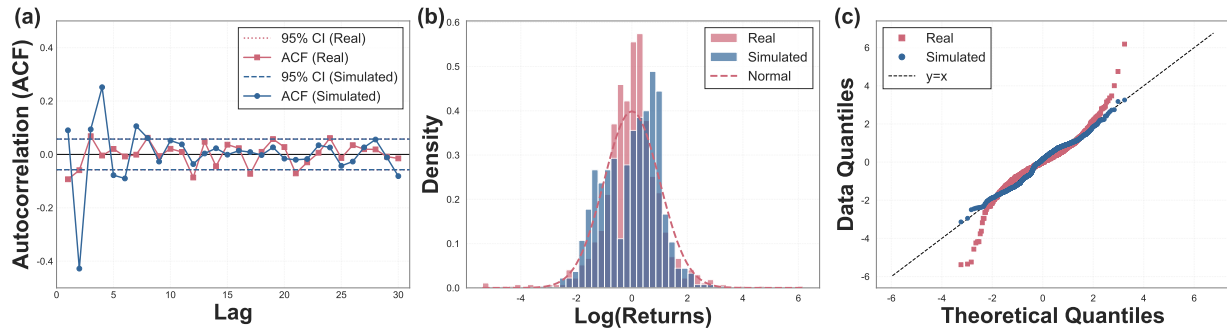


Figure S31. Strategy 1: Stylized facts comparison.

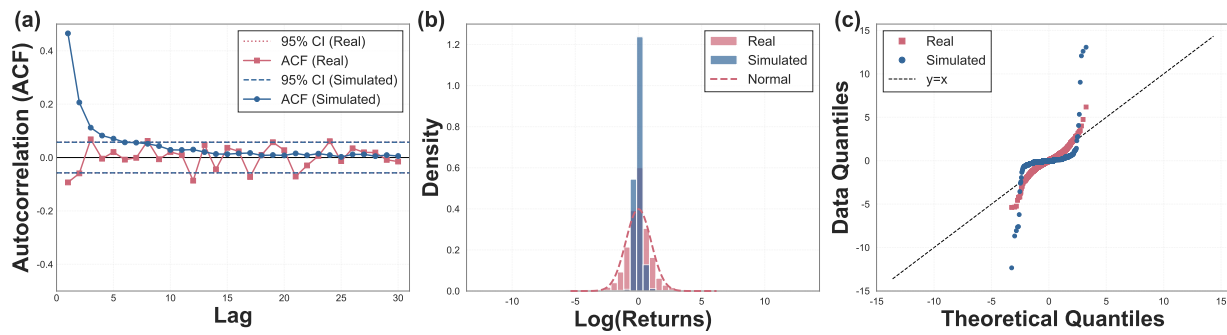


Figure S32. Strategy 2: Stylized facts comparison.

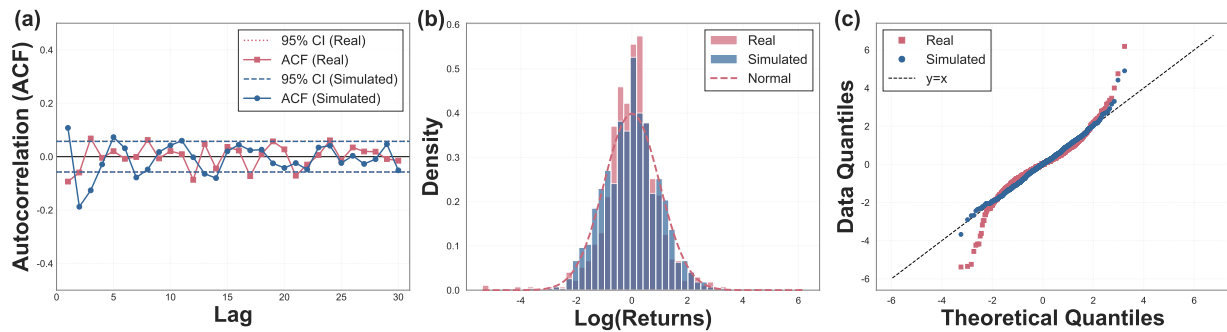


Figure S33. Strategy 3: Stylized facts comparison.

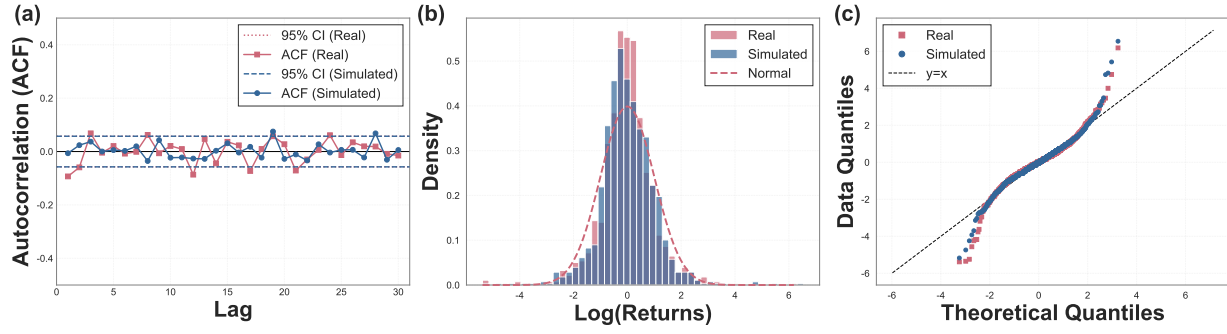


Figure S34. Strategy 4: Stylized facts comparison.

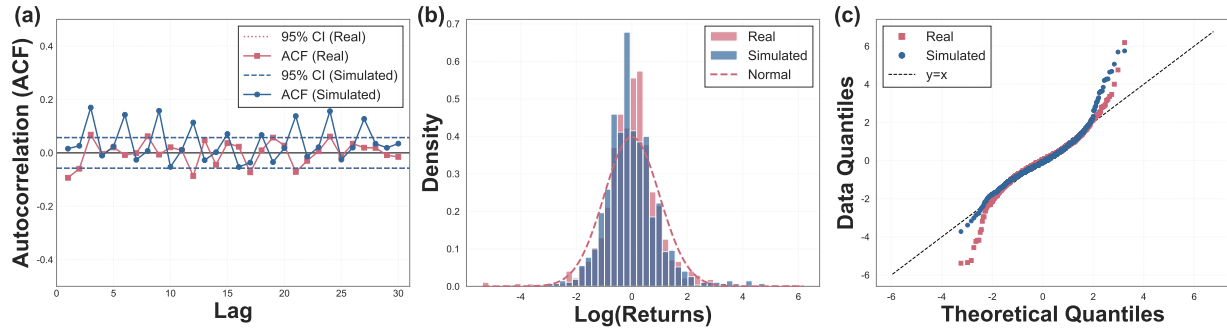


Figure S35. Momentum: Stylized facts comparison.

M. Component Ablation Study

M.1. Price Comparison

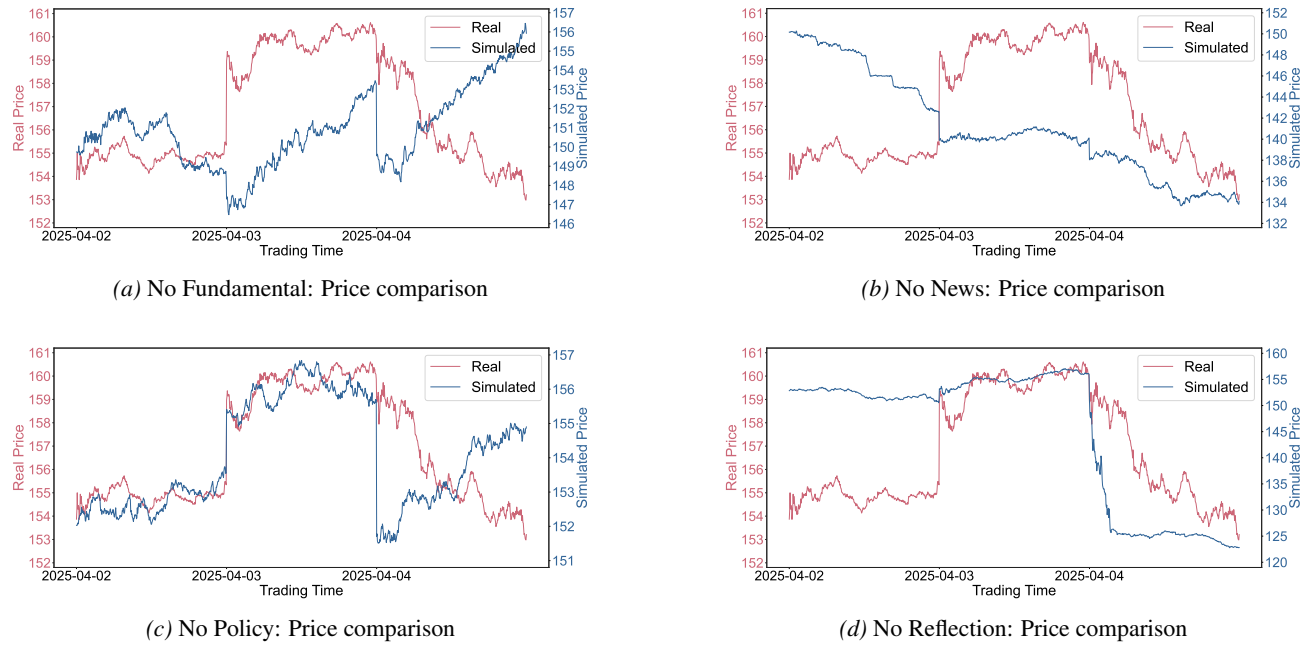


Figure S36. Component ablation study: Price comparisons

M.2. Candlestick Charts

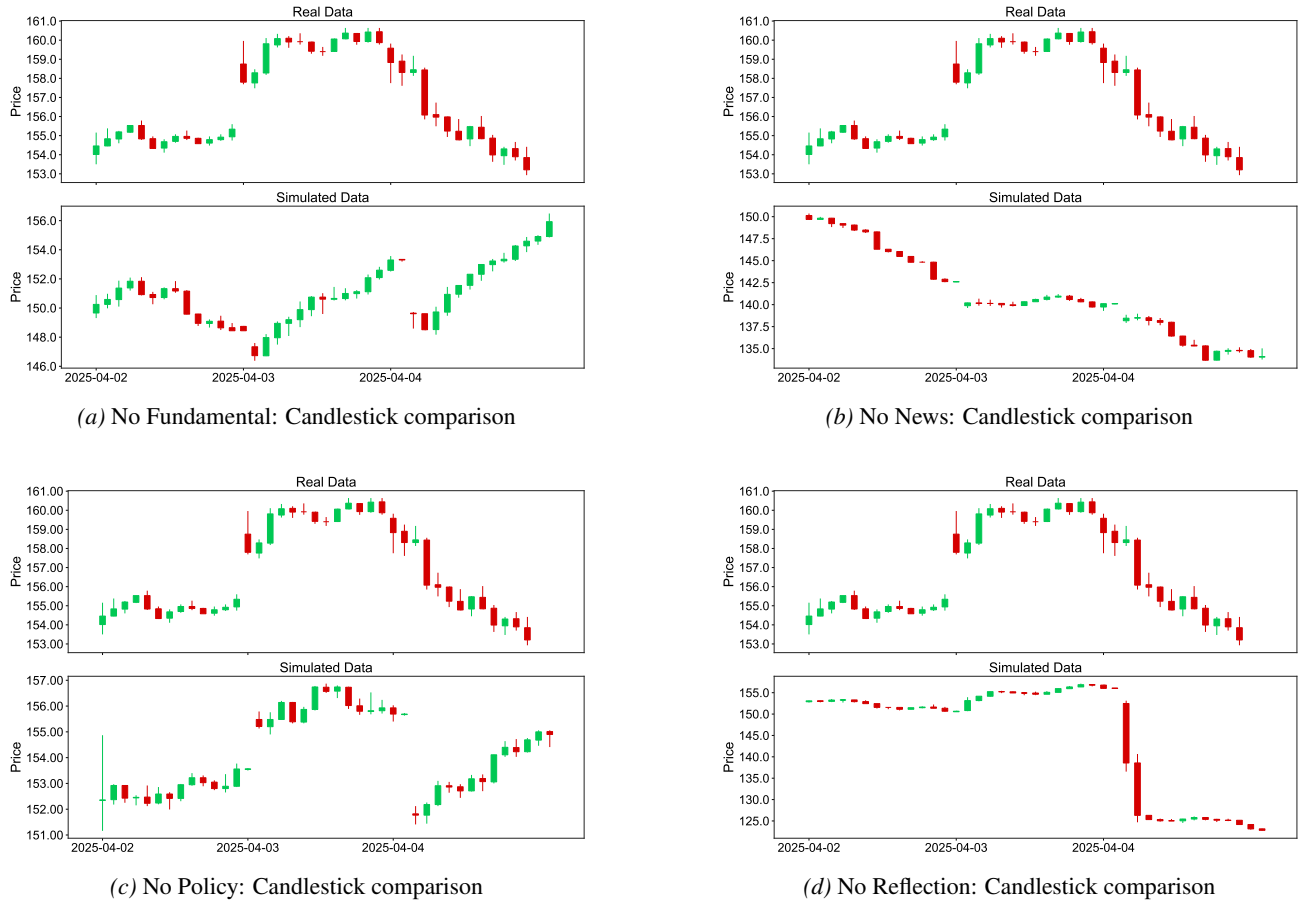


Figure S37. Component ablation study: Candlestick charts comparison

M.3. Stylized facts

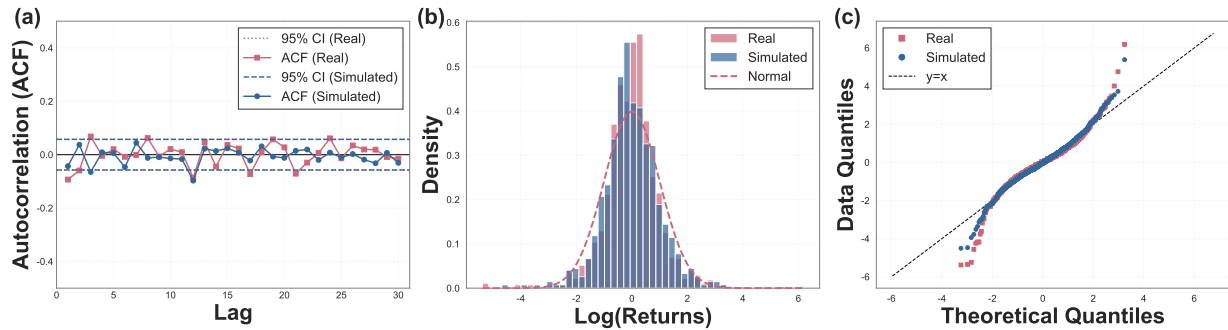


Figure S38. No Fundamental: Stylized facts comparison.

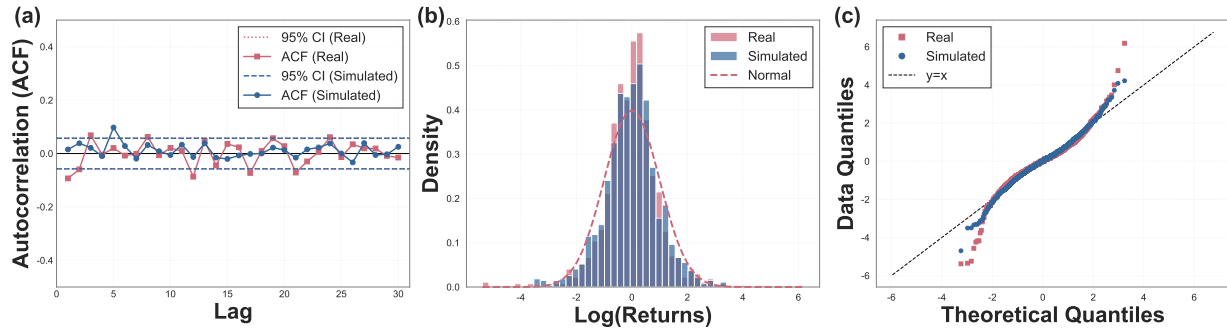


Figure S39. No News: Stylized facts comparison.

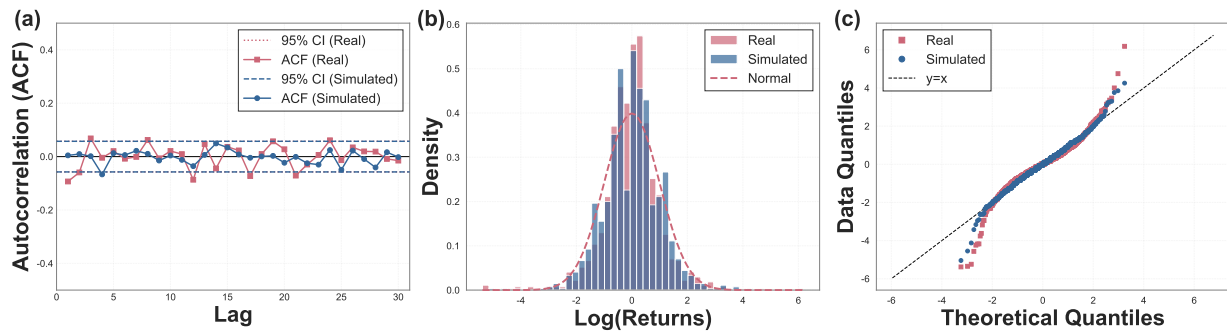


Figure S40. No Policy: Stylized facts comparison.

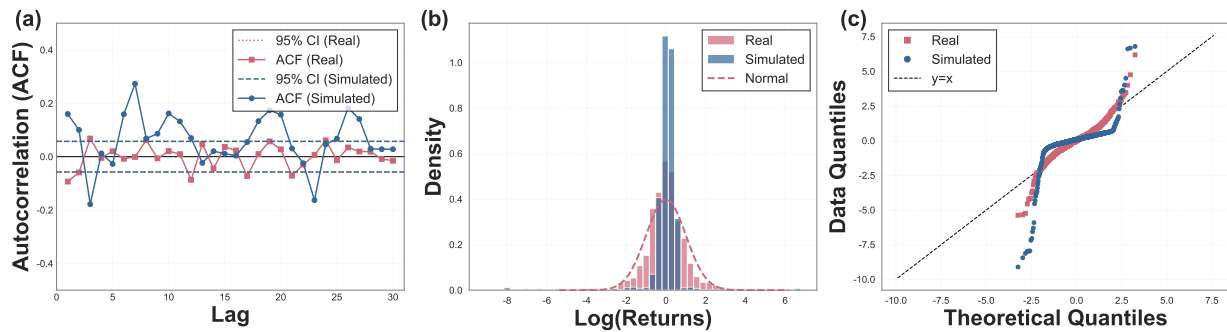
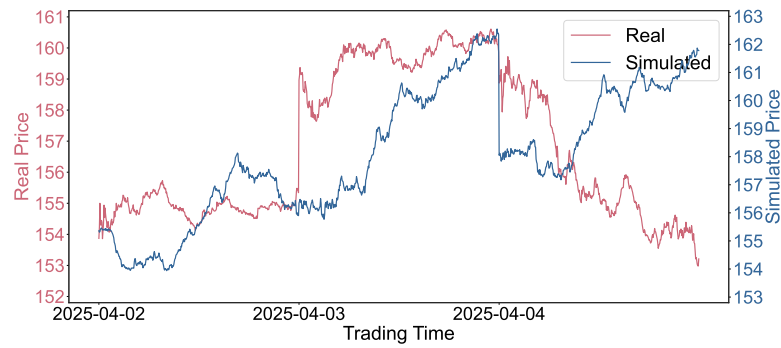


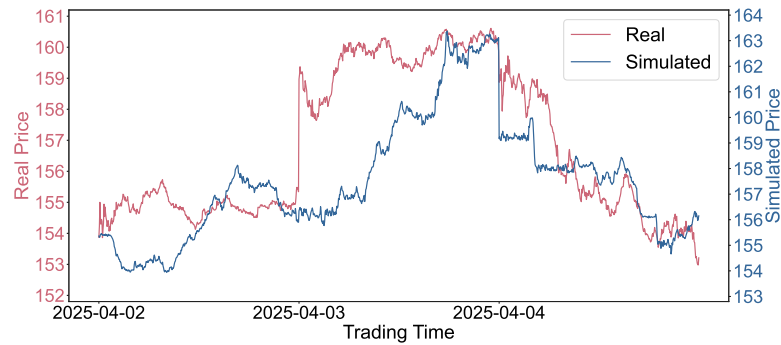
Figure S41. No Reflection: Stylized facts comparison.

N. Liquidity Depletion Shock Analysis

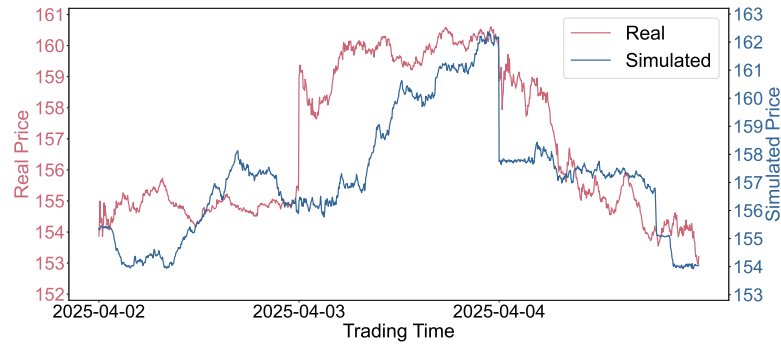
N.1. Price Comparison



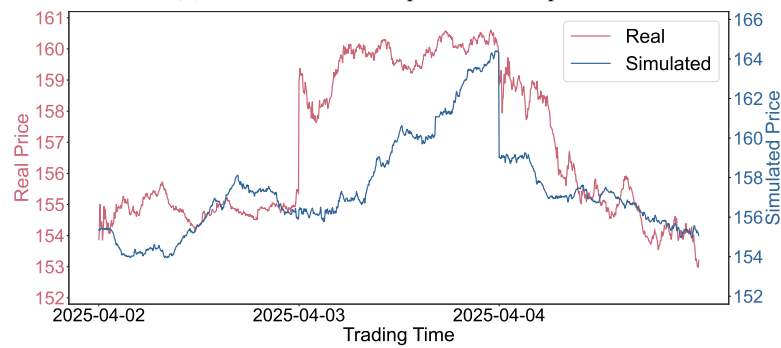
(a) 1% Order Book Depth: Price comparison



(b) 5% Order Book Depth: Price comparison



(c) 50% Order Book Depth: Price comparison



(d) 90% Order Book Depth: Price comparison

Figure S42. Liquidity depletion shock analysis: Price comparisons

N.2. Candlestick Charts

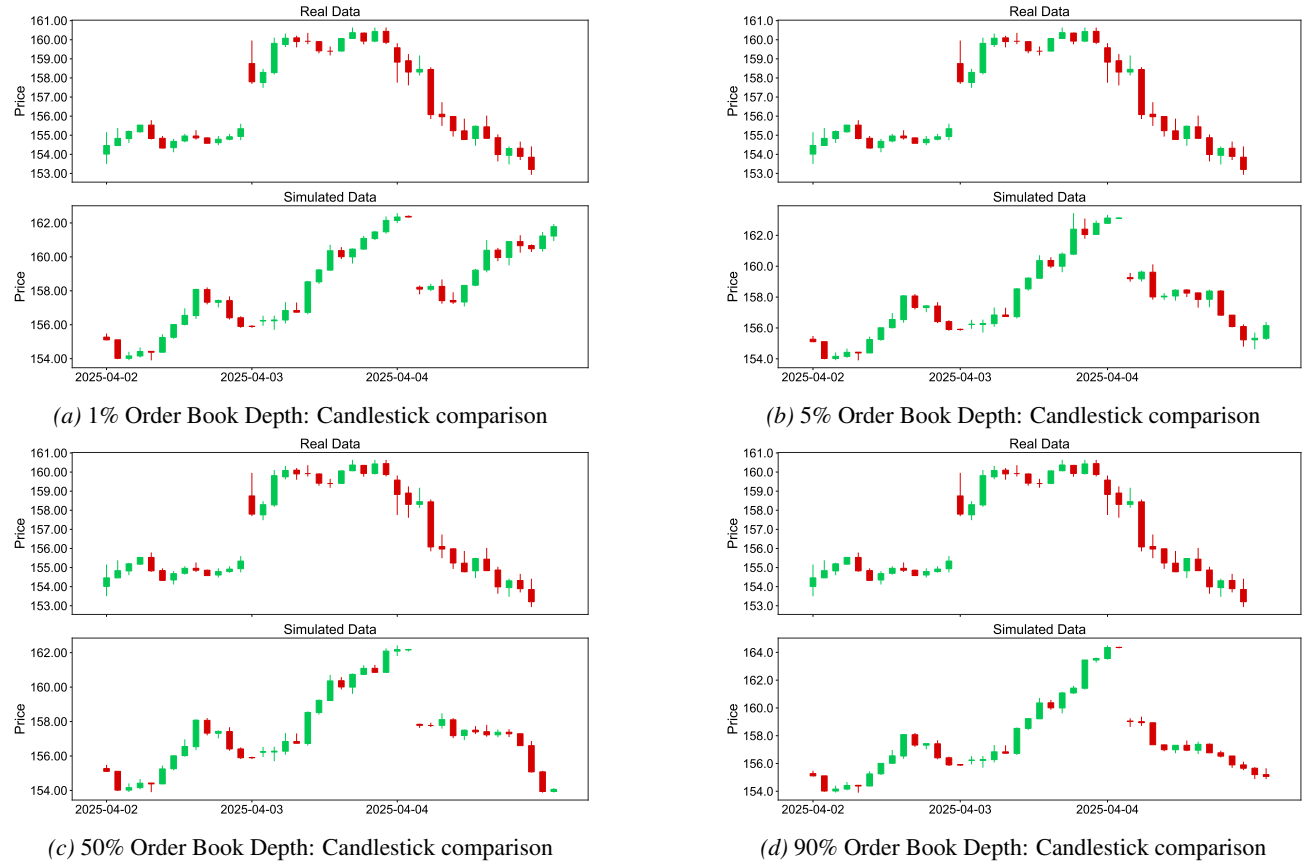


Figure S43. Liquidity depletion shock analysis: Candlestick charts comparison

N.3. Stylized facts

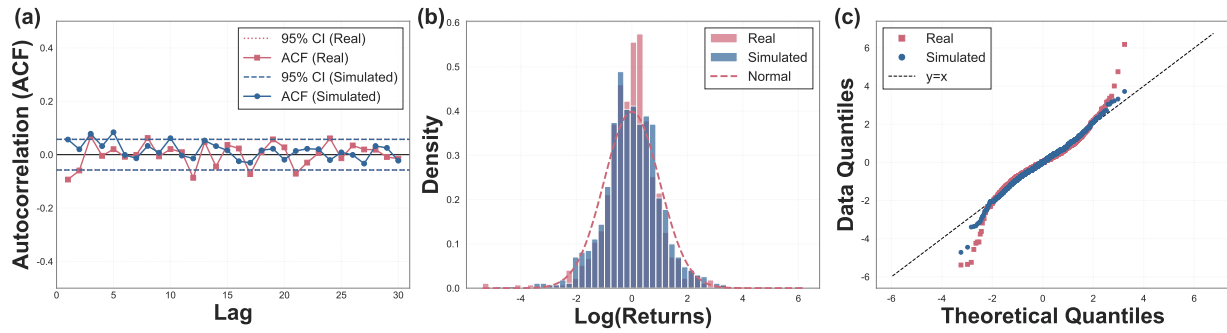


Figure S44. Simulated vs. real JNJ price under 1% Liquidity depletion shock: Stylized facts comparison.

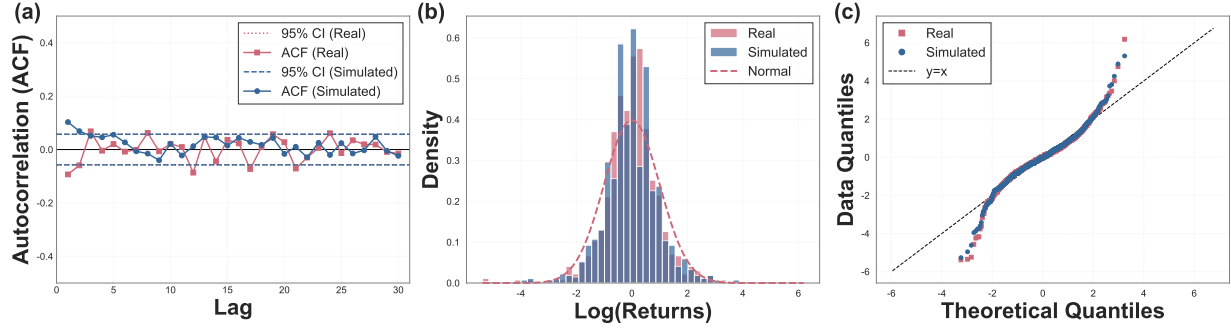


Figure S45. Simulated vs. real JNJ price under 5% Liquidity depletion shock: Stylized facts comparison.

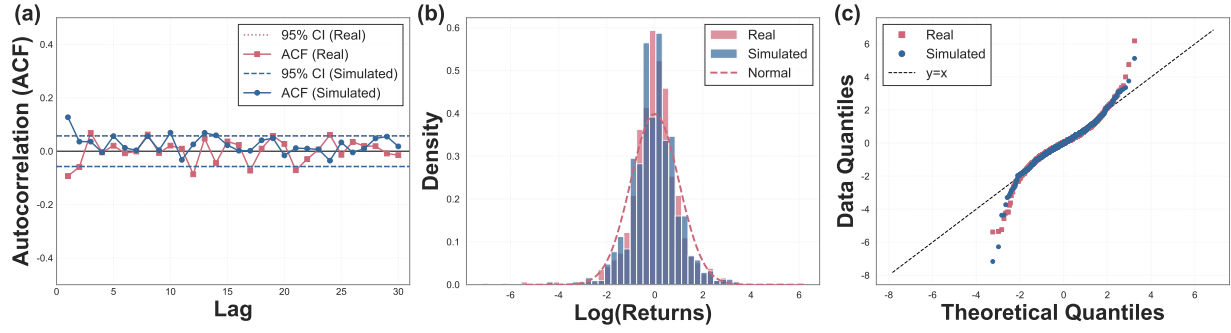


Figure S46. Simulated vs. JNJ price under 50% Liquidity depletion shock: Stylized facts comparison.

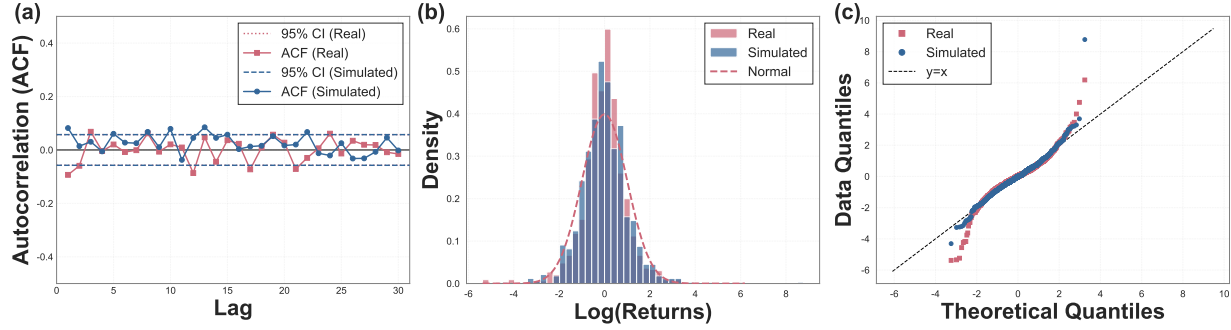


Figure S47. Simulated vs. JNJ price under 90% Liquidity depletion shock: Stylized facts comparison.

O. Price Discovery Process

MarketSim does not use any external price inputs or anchor to any reference price. All prices emerge endogenously through a standard continuous double auction (CDA) identical to those used in real-world equity markets. Buy and sell limit orders arrive asynchronously and populate the order book

$$\mathcal{B}_t = \{(p_i^b, q_i^b)\}, \quad \mathcal{S}_t = \{(p_j^s, q_j^s)\}.$$

The best bid and best ask are defined as

$$\text{BestBid}_t = \max_i p_i^b, \quad \text{BestAsk}_t = \min_j p_j^s.$$

A transaction occurs when a marketable order arrives and the crossing condition

$$\text{BestBid}_t \geq \text{BestAsk}_t$$

is satisfied. The execution price follows the standard counterparty-price rule:

$$P_t = \begin{cases} \text{BestAsk}_t & \text{if a buy order crosses the book,} \\ \text{BestBid}_t & \text{if a sell order crosses the book.} \end{cases}$$

The executed volume is

$$Q_t = \min(q_{\text{best}}^b, q_{\text{best}}^s),$$

and the corresponding order quantities are updated.

Because P_t depends solely on the internal liquidity state (B_t, S_t) , the simulator has no access to real-world prices or future information. All market dynamics arise from the endogenous interactions of heterogeneous agents operating within the CDA mechanism, ensuring that the resulting price series reflect emergent behavior rather than any form of data leakage or externally imposed reference price.

P. Computational Details

All simulations run efficiently on standard CPU hardware. The experiments operate on an Intel(R) Xeon(R) Platinum 8378A CPU at 3.00GHz, without requiring any local GPU clusters. The event-driven architecture maintains a low computational load, with an average usage of 4.07 CPU cores and a peak of 11.0 cores during initialization. Once stabilized, the system operates between 2.45 and 3.20 cores, exhibiting consistent throughput.

For agent reasoning, MarketSim uses the DeepSeek API to balance performance and computational cost. A typical experiment consumes approximately 3.98 million input tokens and 0.40 million output tokens. Across the simulation horizon, the system processes 7.62 million limit orders and completes the run in approximately 4.5 hours, demonstrating that MarketSim scales to high-frequency trading workloads while remaining computationally lightweight.

Q. Reproducibility Statement

To ensure the reproducibility of our results, all codes for MarketSim framework are available at <https://anonymous.4open.science/r/MarketSim-E854/>. Our primary large language model is DeepSeek R1, with Qwen3-8b and Llama-3.1-8b used for generalizability experiments. The composition of the agent population in our simulation is designed to mirror the participant structure of the real-world NASDAQ market (Brav et al., 2024; Blume et al., 2017) and prior practices (Byrd et al., 2020). Please check more details (e.g., specific prompts and computational resource) in Appendix.

R. Ethics Statement

No human participants are involved in this study, and no ethical issues are applicable.

S. Use of Large Language Models

After completing the initial draft, we use LLMs to polish the text and consult them on specific word choices.